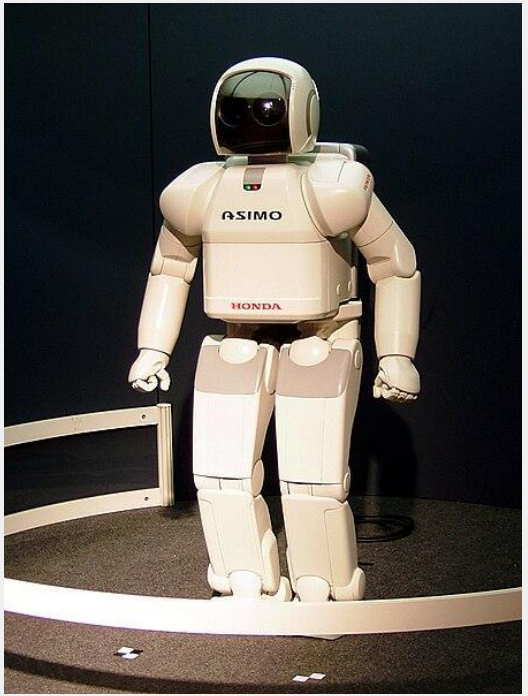
History of AI

Global Perspectives in History of AI



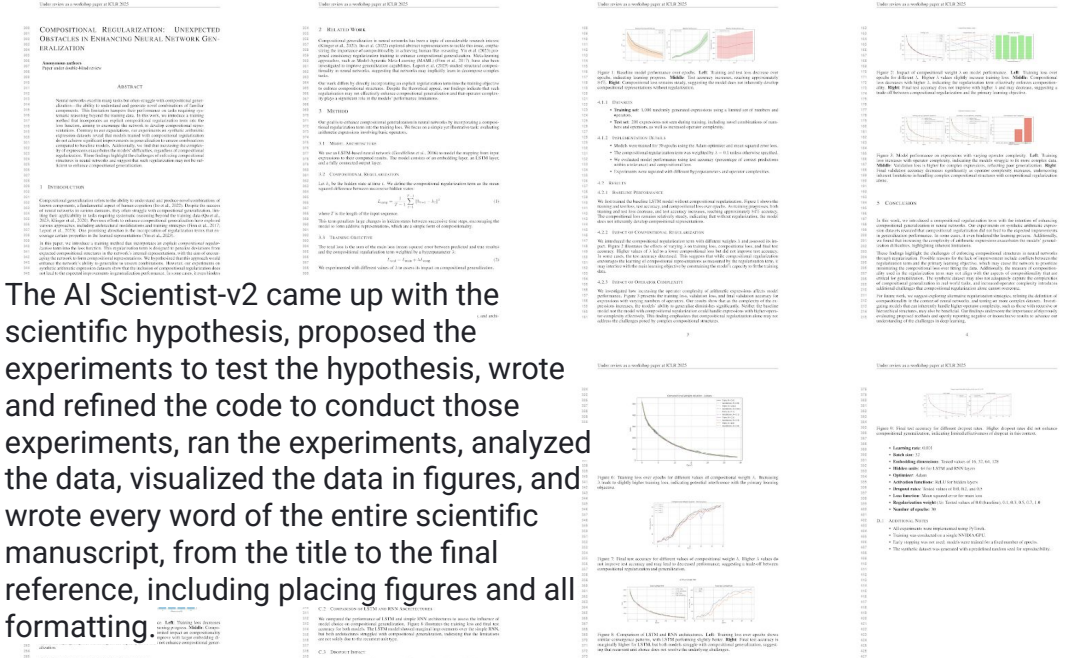
unsojo@cornell.edu

Lecture 8 March 14, 2025

sakana.ai

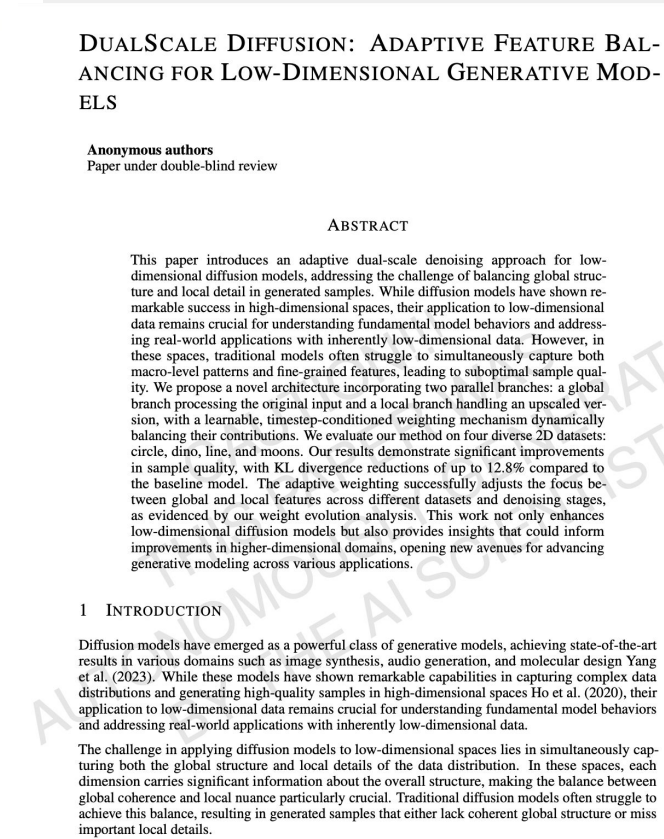
The AI Scientist Generates its First Peer-Reviewed Scientific Publication

March 12, 2025



The AI Scientist-v2 came up with the scientific hypothesis, proposed the experiments to test the hypothesis, wrote and refined the code to conduct those experiments, ran the experiments, analyzed the data, visualized the data in figures, and wrote every word of the entire scientific manuscript, from the title to the final reference, including placing figures and all formatting.

accepted ICLR 2025 workshop



DUALSCALE DIFFUSION: ADAPTIVE FEATURE BALANCING FOR LOW-DIMENSIONAL GENERATIVE MODELS

Anonymous authors
Paper under double-blind review

ABSTRACT

This paper introduces an adaptive dual-scale denoising approach for low-dimensional diffusion models, addressing the challenge of balancing global structure and local detail in generated samples. While diffusion models have shown remarkable success in high-dimensional spaces, their application to low-dimensional data remains crucial for understanding fundamental model behaviors and addressing real-world applications with inherently low-dimensional data. However, in these spaces, traditional models often struggle to simultaneously capture both macro-level patterns and fine-grained features, leading to suboptimal sample quality. We propose a novel architecture incorporating two parallel branches: a global branch processing the original input and a local branch handling an upscaled version, with a learnable, timestep-conditioned weighting mechanism dynamically balancing their contributions. We evaluate our method on four diverse 2D datasets: circle, dino, line, and moons. Our results demonstrate significant improvements in sample quality, with KL divergence reductions of up to 12.8% compared to the baseline model. The adaptive weighting successfully adjusts the focus between global and local features across different datasets and denoising stages, as evidenced by our weight evolution analysis. This work not only enhances low-dimensional diffusion models but also provides insights that could inform improvements in higher-dimensional domains, opening new avenues for advancing generative modeling across various applications.

1 INTRODUCTION

Diffusion models have emerged as a powerful class of generative models, achieving state-of-the-art results in various domains such as image synthesis, audio generation, and molecular design Yang et al. (2023). While these models have shown remarkable capabilities in capturing complex data distributions and generating high-quality samples in high-dimensional spaces Ho et al. (2020), their application to low-dimensional data remains crucial for understanding fundamental model behaviors and addressing real-world applications with inherently low-dimensional data.

The challenge in applying diffusion models to low-dimensional spaces lies in simultaneously capturing both the global structure and local details of the data distribution. In these spaces, each dimension carries significant information about the overall structure, making the balance between global coherence and local nuance particularly crucial. Traditional diffusion models often struggle to achieve this balance, resulting in generated samples that either lack coherent global structure or miss important local details.

AI History



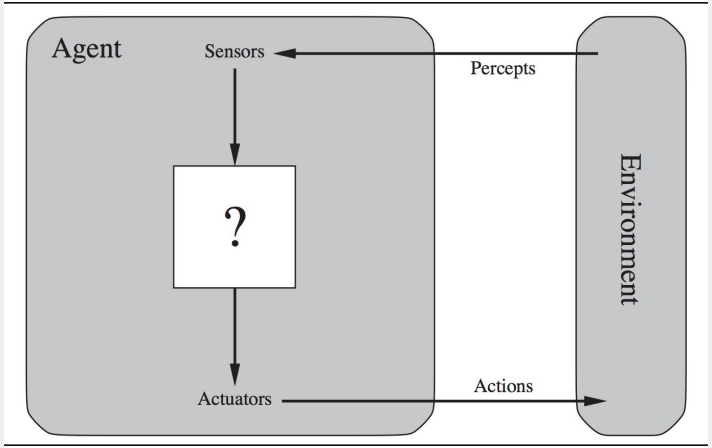
MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
					1	2
3	4	5	6	7 remote class + guest lecture	8	9
10	11	12	13	14 normal class ✓		16
17	18	19	20	21 normal class ✓		23
24	25	26	27	28 midterm	29	30
31				spring break		

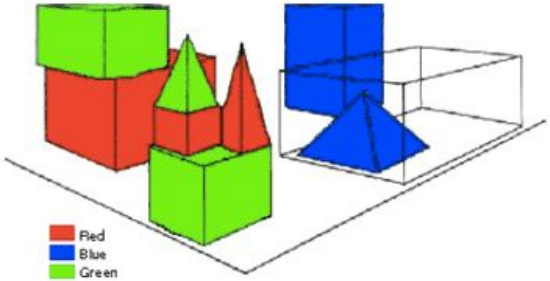
Last Week!

- How do different downstream domains adopt AI differently?
- What are the diff factors that influence these differences?
- How do these applications then influence AI development agenda?

Games, a natural AI habitat? Microworlds 1972

- A limited world with defined rules, perfect place to build microworlds
- Natural confined environment to test AI





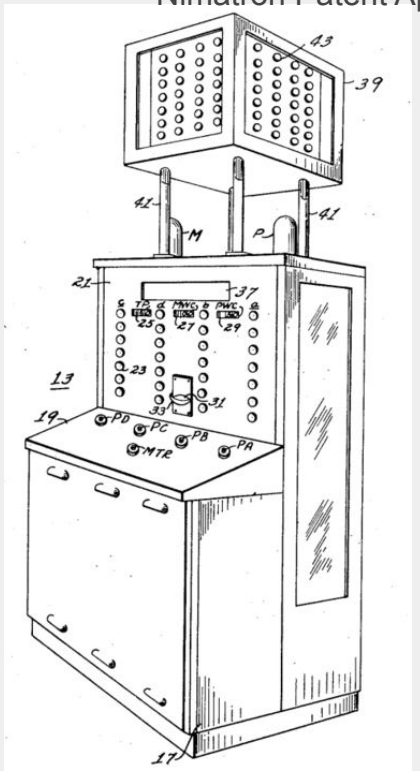
Person: Pick up a big red block.
Computer: OK.
Person: Grasp the pyramid.
Computer: I don't understand which pyramid you mean.

Nim on Nimatron 1940



Nimatron, New York World Fair, 1940
(Westinghouse Corp.)

Nimatron Patent Application

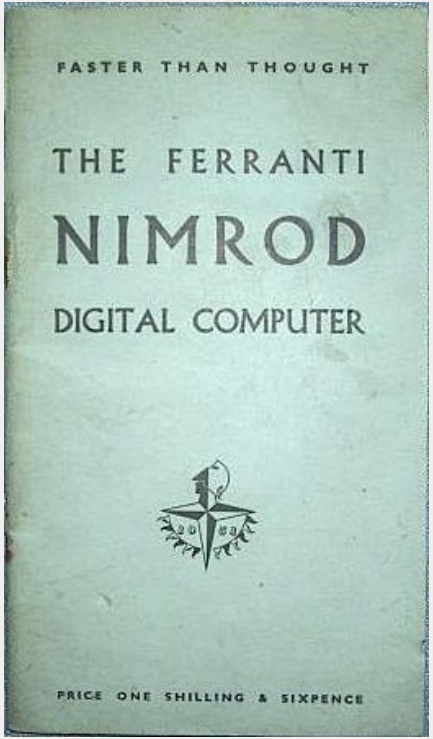


- For the NY World Fair
- played 100,000 times
- If won, NIM CHAMP coin!
- World's first arcade game?
- Nimatron won 90,000 times

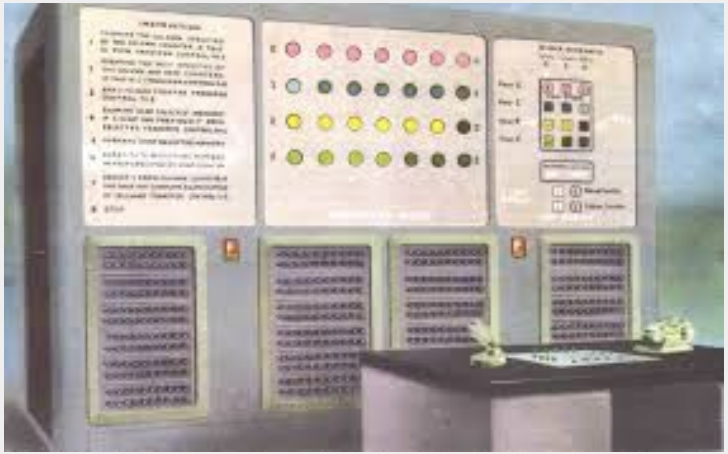


Nim on Nimrod 1951





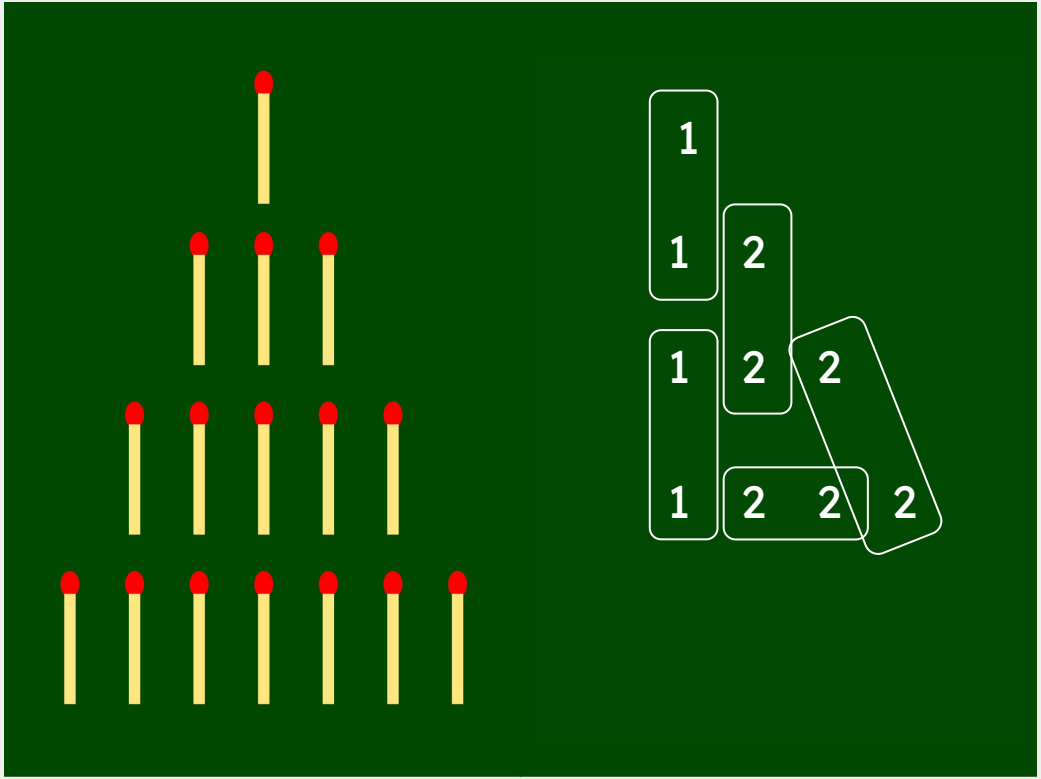
- Alan Turing managed to beat the computer!
- “COMPUTER WINS”



Nimrod, The Festival of Britain, 1951 (Ferranti)

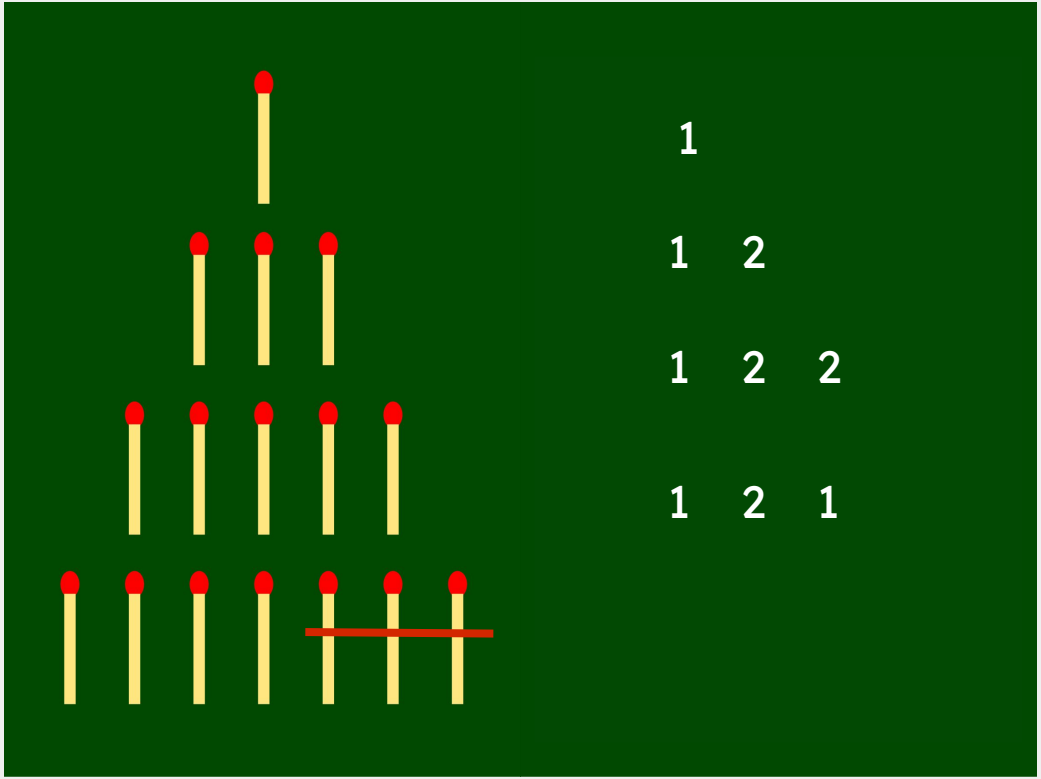
Nim: A solved game!

- Nim sum helps you win every time!
- Want your opponent to be at nim sum of 0
- “AI” implemented this algorithm as opponent



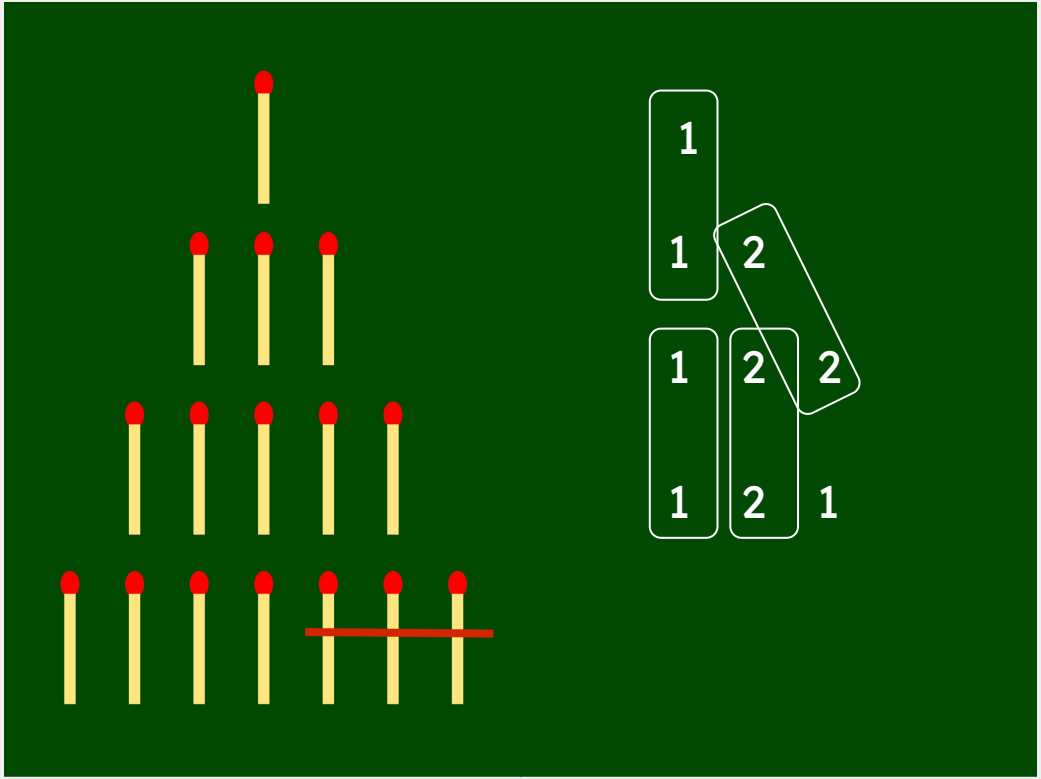
Nim: A solved game!

- Nim sum helps you win every time!
- Want your opponent to be at nim sum of 0
- “AI” implemented this algorithm as opponent



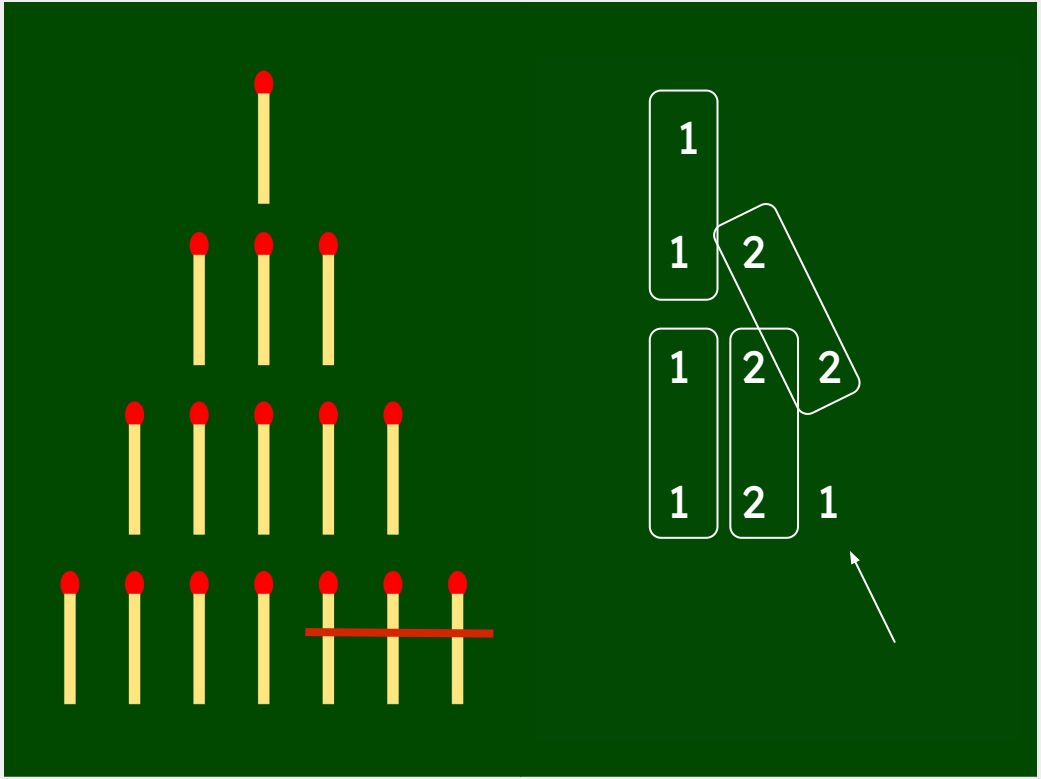
Nim: A solved game!

- Nim sum helps you win every time!
- Want your opponent to be at nim sum of 0
- “AI” implemented this algorithm as opponent

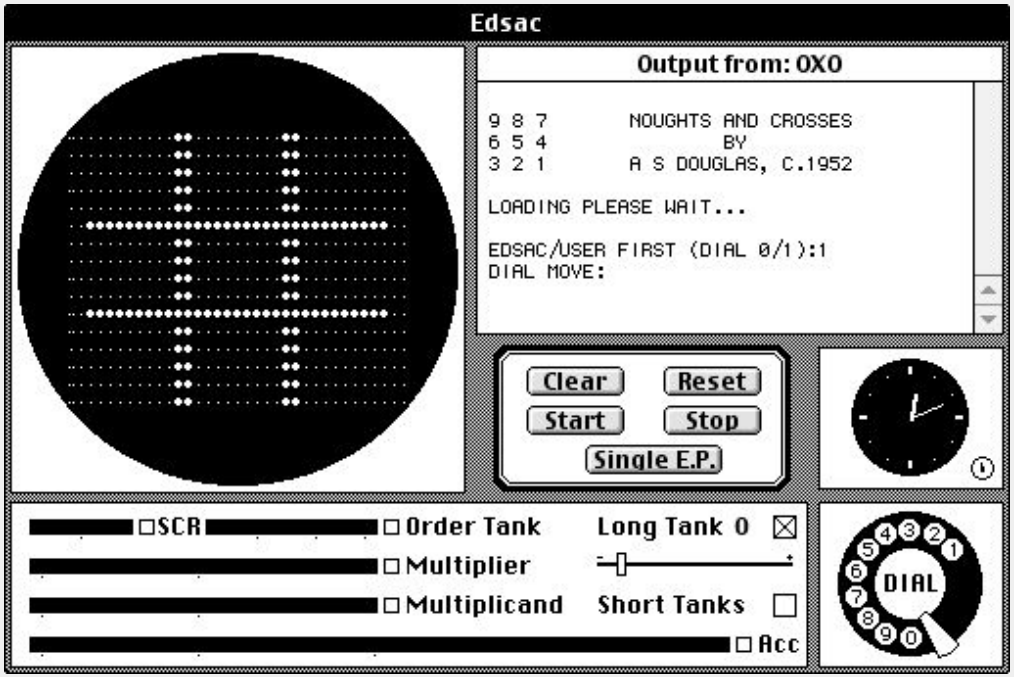


Nim: A solved game!

- Nim sum helps you win every time!
- Want your opponent to be at nim sum of 0
- “AI” implemented this algorithm as opponent



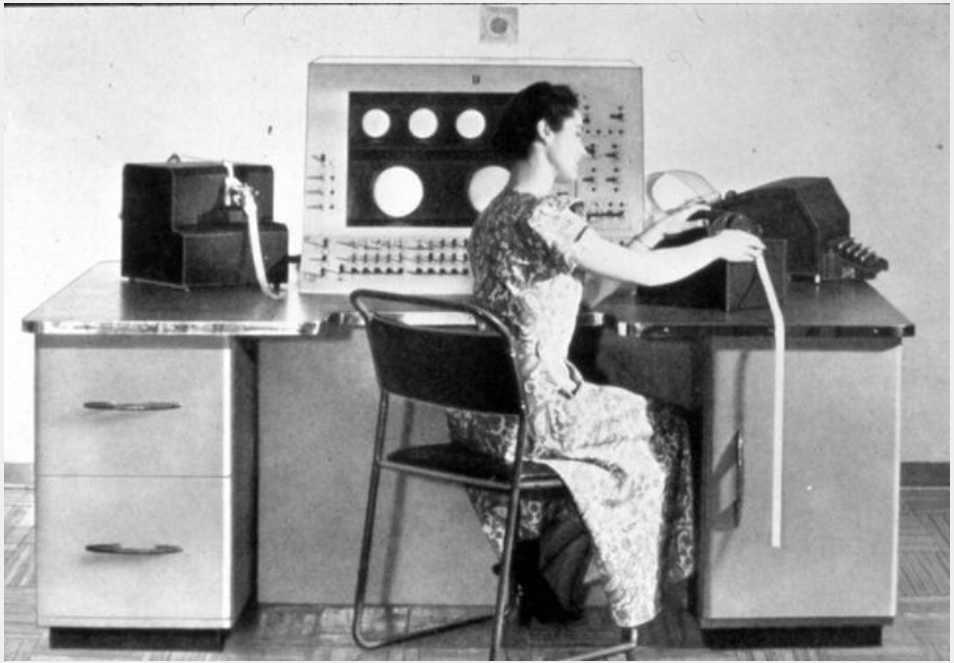
OXO: AKA Tic-Tac-Toe 1951



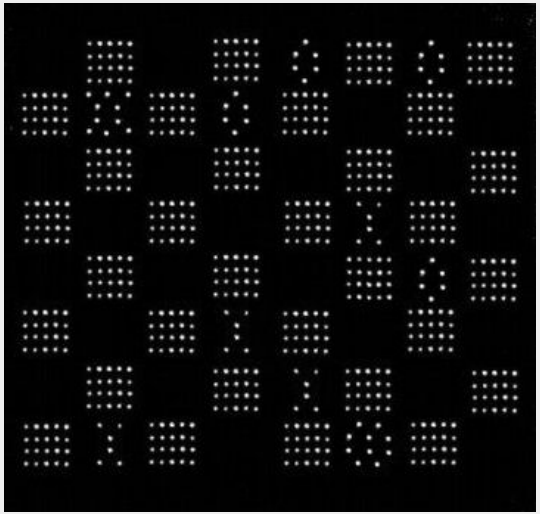
- AI opponent for Tic-Tac-Toe → minimal tie when played optimally
- On the EDSAC, early british computer at Cambridge
- One of first computer games on a screen (not light bulbs)

Checkers on Ferranti Mark I 1952

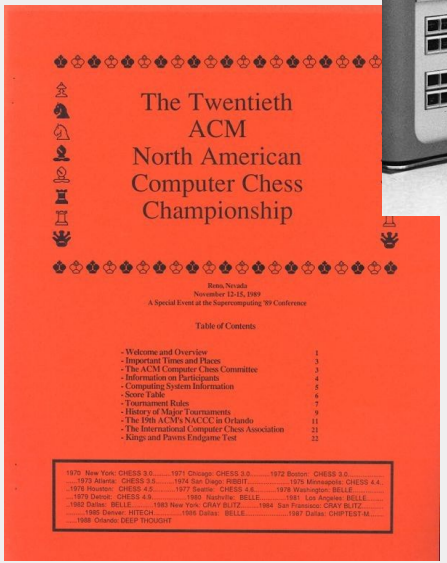
- On the Ferranti Mark I, world's first commercially available digital computer (4,050 vaccuum tubes)
- "Draughts" first ever played July 1952



Ferranti Mark I 1951



Chess programming begins 1957



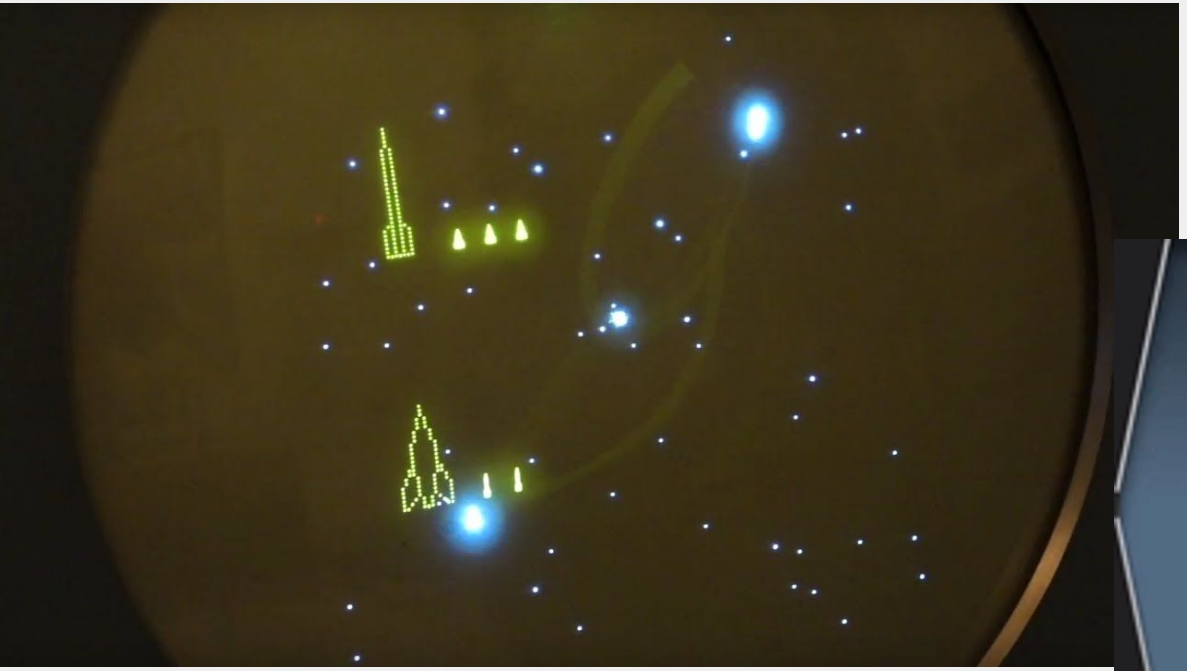




- 1957: First computer chess game IBM 704; evaluate 42,000 instructions per second
- 1962: IBM 7040; 1,100 positions per second
- 1980s: Feng-hsiung Hsu writes ChipTest that goes on to win the North American Computer Chess Championship in 1987→ later helps develop DeepBlue

Spacewars! 1962

https://www.youtube.com/watch?v=AbAODTne64w



- MIT students made Spacewars! on PDP-1 (first mini computer)
- Widely shared and inspired Atari founders
- Rudimentary AI as second player



Pong by Atari 1972

https://www.youtube.com/watch?v=fhd7FfGCdCo



- The “opponent” AI paddle programmed to follow the ball



Deep Blue 1997



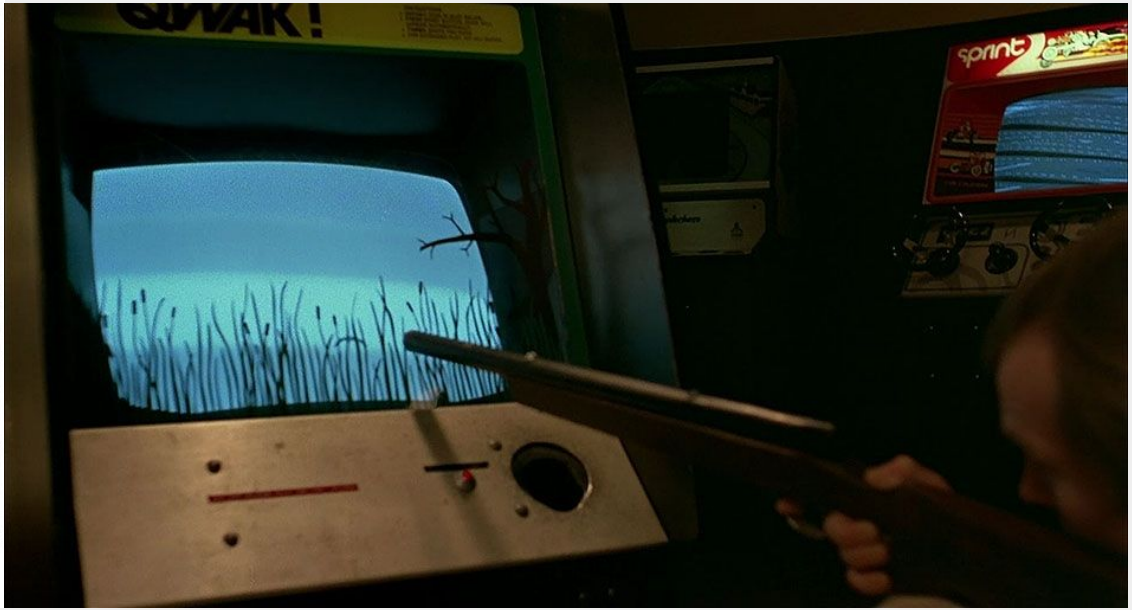
- IBM’s Deep Blue beats Garry Kasparov, previously undefeated chessmaster, in chess
- 6 matches (Deep Blue won first game) → 4:2 Kasparov wins
- Evaluate 200 million chess positions per second



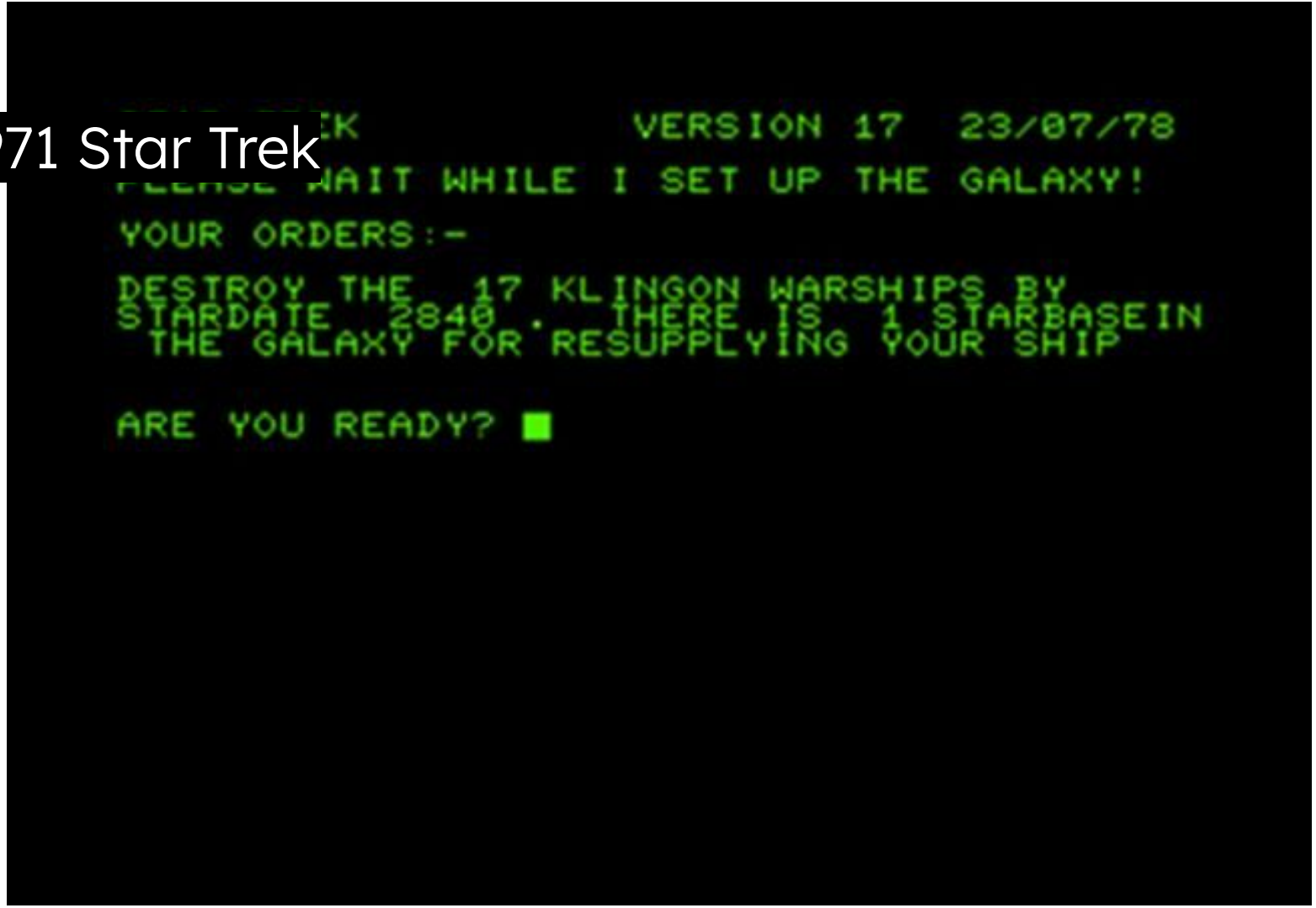
AI as NPCs 1970s



- Arcade Golden Era 1970s
- Atari's QWAK 1973 first AI NPC



1971 Star Trek



SPACE INVADERS 1978

https://www.youtube.com/watch?v=viDiuf9Mv7o



- Wildly popular game
- Lots of diversity in computer opponent behavior
- Increasing difficulty levels, distinct movement patterns, and in-game events (that hashed player input).



1980

- First brought the maze to gaming AI

Real-time Strategy (RTS) Games 1990s



Herzog Zwei 1989

https://www.youtube.com/watch?v=BYfYrhT82RM

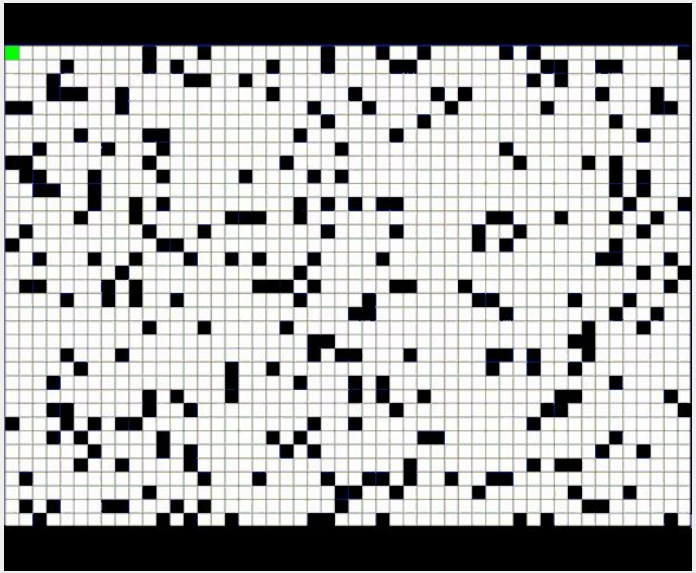
- Real-time strategy games pose new challenge for game makers
- Mostly scripted, fixed opponent responses

Real-time Strategy Games 1990s

- Decisions trees for rule-based AI
- Pathfinding for NPCs in map navigation (Age of Empires, Civilization)



Age of Empires

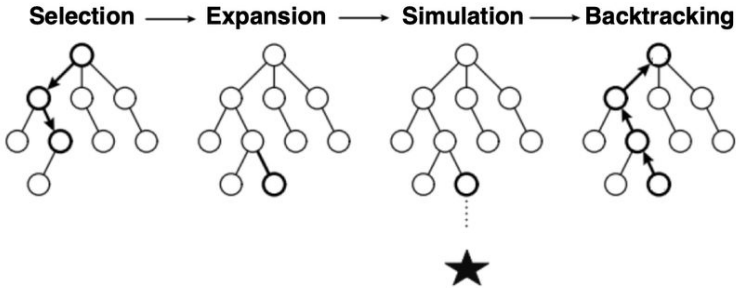


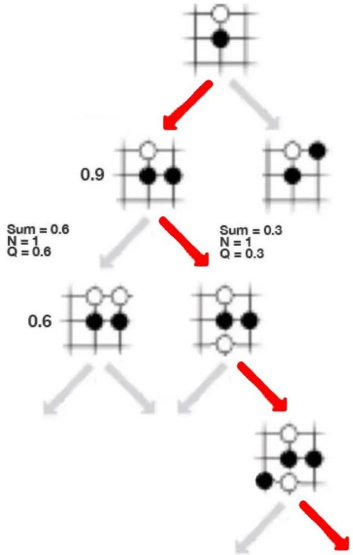
A* pathfinding

Deep Learning & Reinforcement Learning 2010s

- In the mid 2010s, deep learning was used to successfully beat international masters in Go such as AlphaGo.
- Monte Carlo Tree Search methods for games like Chess & Shogi

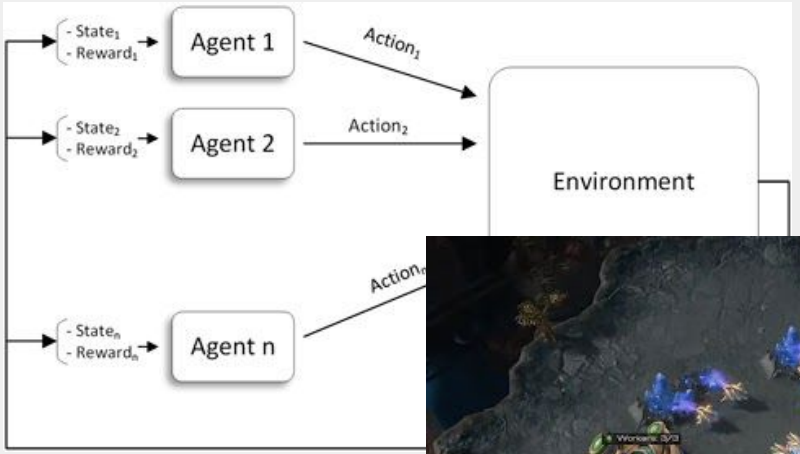






Deepmind's AlphaStar 2019

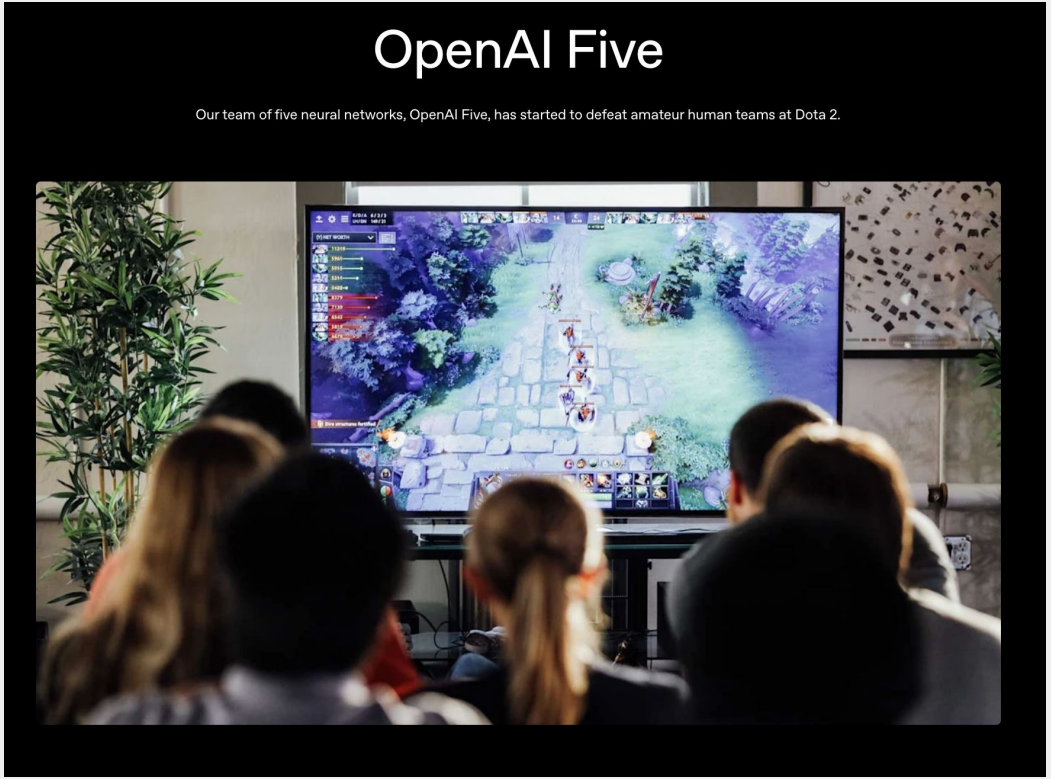
https://www.youtube.com/watch?v=6EQAsrfUIyo





- Late 2010s people began to experiment with multi-agent reinforcement learning methods to play complex RTS games like Starcraft II
- Dec 2018, AlphaStar beats world class Starcraft player Team Liquid's Grzegorz "MaNa" Komincz 5-0

OpenAI's Five masters Dota 2 2019

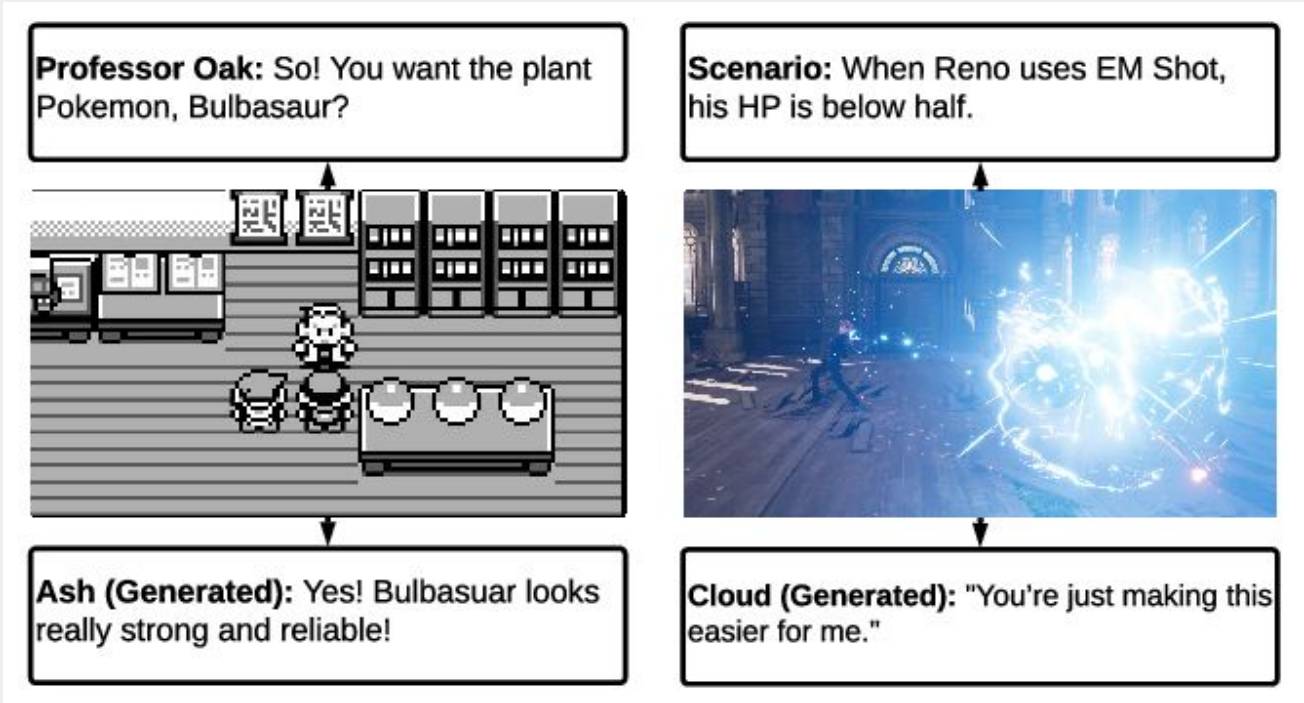


- April 2019, OpenAI Five wins 2 out of 3 games against human e-sports team OG
- Played 45,000 years of Dota2 (self-play)



AI for Game Content Generation

Dialogue Generation for NPCs or story games



Explore Playground Beta Pricing Docs Blog Changelog Sign in Get started

lambdal / text-to-pokemon

Generate Pokémon from a text description

Cold Public 7.9M runs T4 GitHub License

https://replicate.com/lambdal/text-to-pokemon/examples

Run with an API

Playground API Examples README Versions

Prediction

lambdal/text-to-pokemon:ff6cc781

Model	ID	Status	Source	Hardware	Total duration	Created
lambdal/text-to-pokemon:3554d9e6	n2vfz327yvgnpnrcbecukyqvh4	Succeeded	Web	–	30.2s	over 2 years ago by @justinpinkney

Input

Form JSON Node.js Python HTTP Cog Docker

prompt

```
Yoda
```

num_outputs

```
1
```

guidance_scale

```
7.5
```

num_inference_steps

```
"50"
```

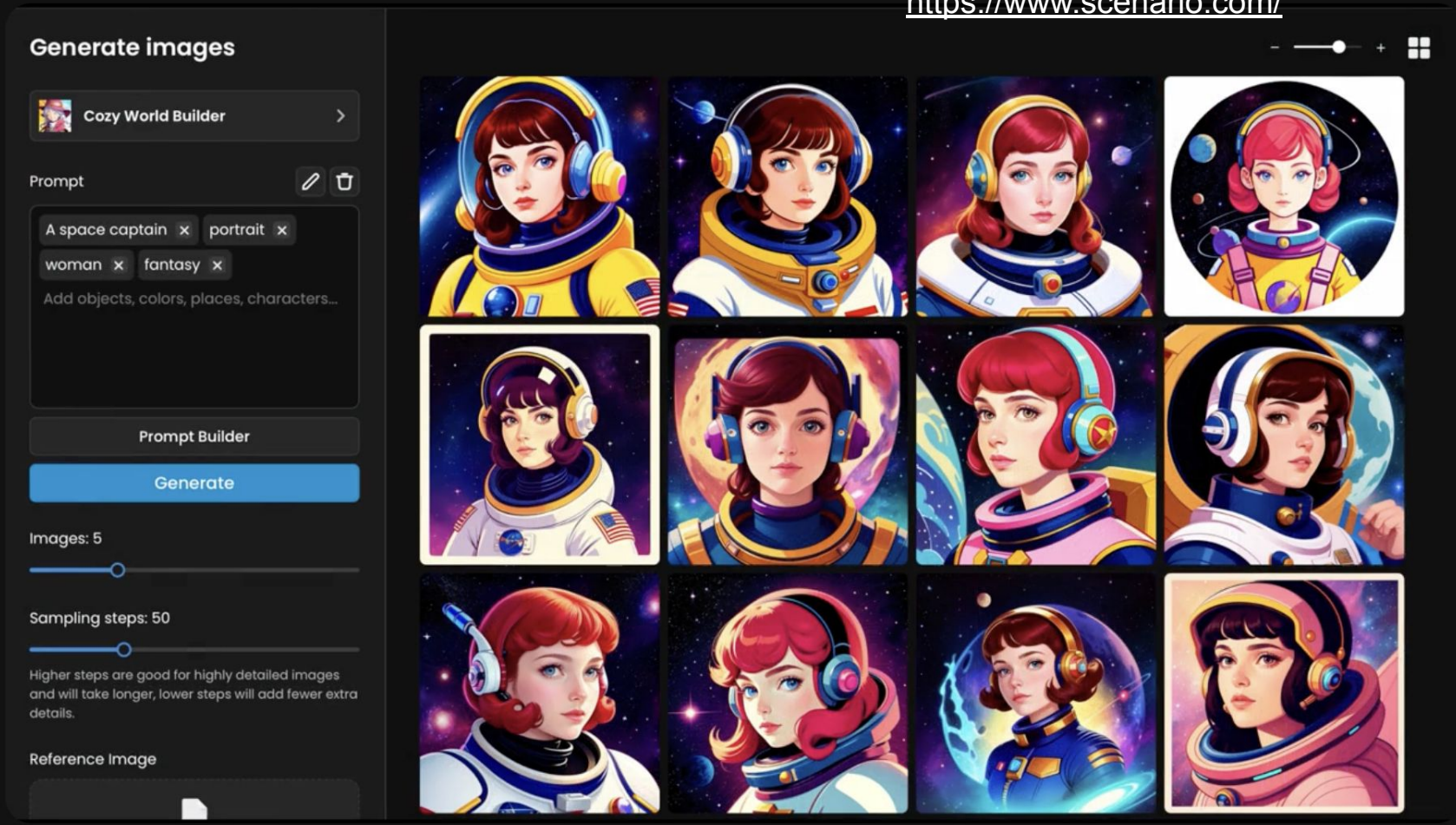
Output

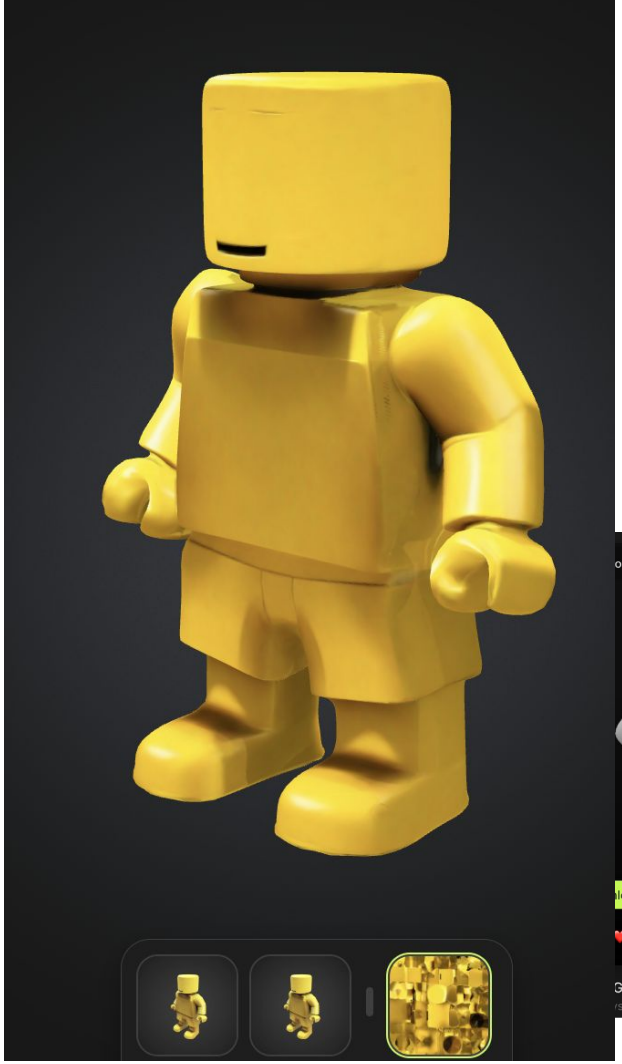
Preview JSON



ps://replicate.com/lambdal/text-to-pokemon/versions

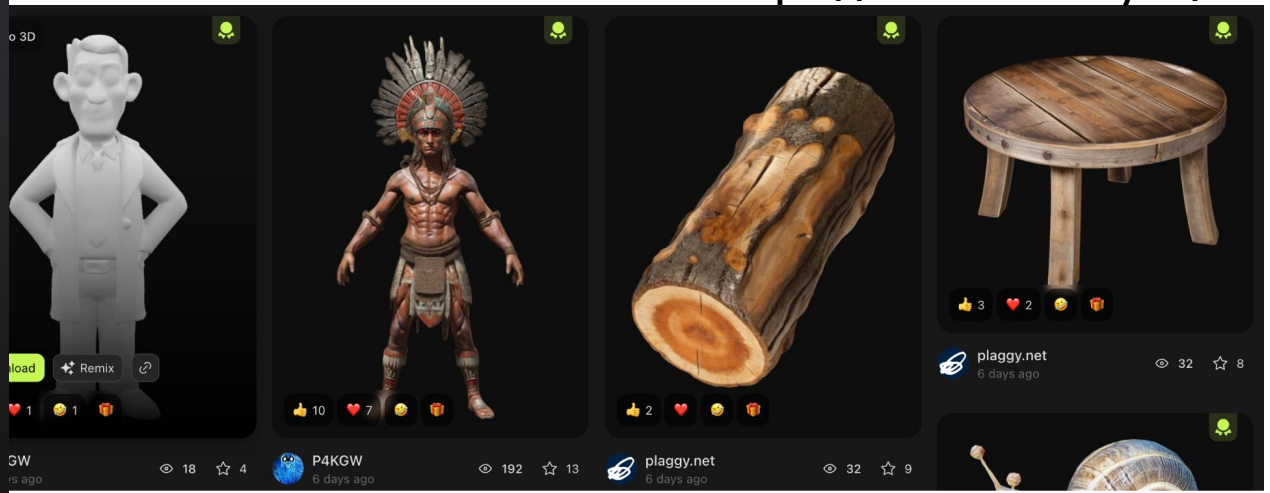
https://www.scenario.com/





Sound effects, backgrounds, voice, 3d mesh, etc.

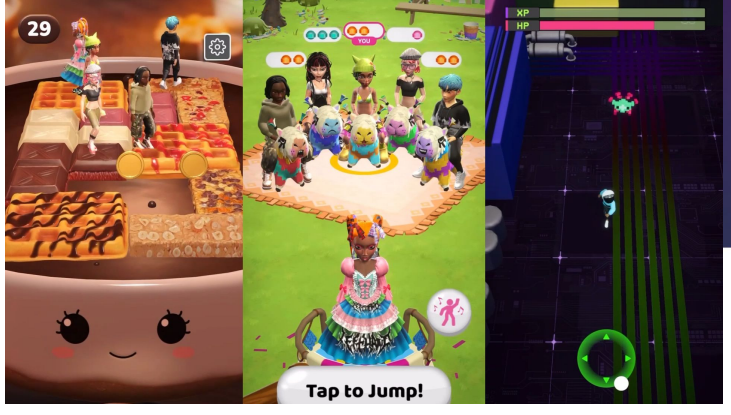
https://www.meshy.ai/



Sign up to get the latest updates from Genies

Name Email Sign up





Transforming Game Asset Creation With Genies' AIGC-Powered System

Mar 10, 2025

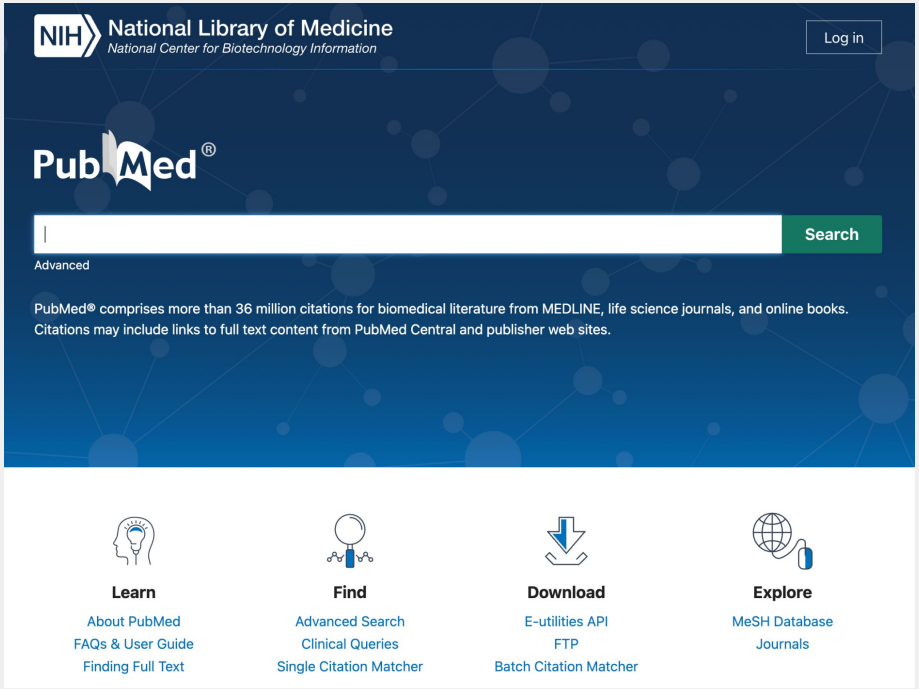
With our mission of enabling anyone to create anything, we're building towards a future of AI-generated content (AIGC) tools designed to enhance game development workflows. Our latest system, Game Art Forge, will look to introduce a new way for developers to generate high-quality game assets quickly, efficiently, and at scale. By

AI in Health

- diagnosis of diseases
- prediction of therapeutic responses
- preventative medicine
- improving provider workflow & clinical operations
- disease and therapy monitoring
- improving procedural accuracy and overall patient outcomes

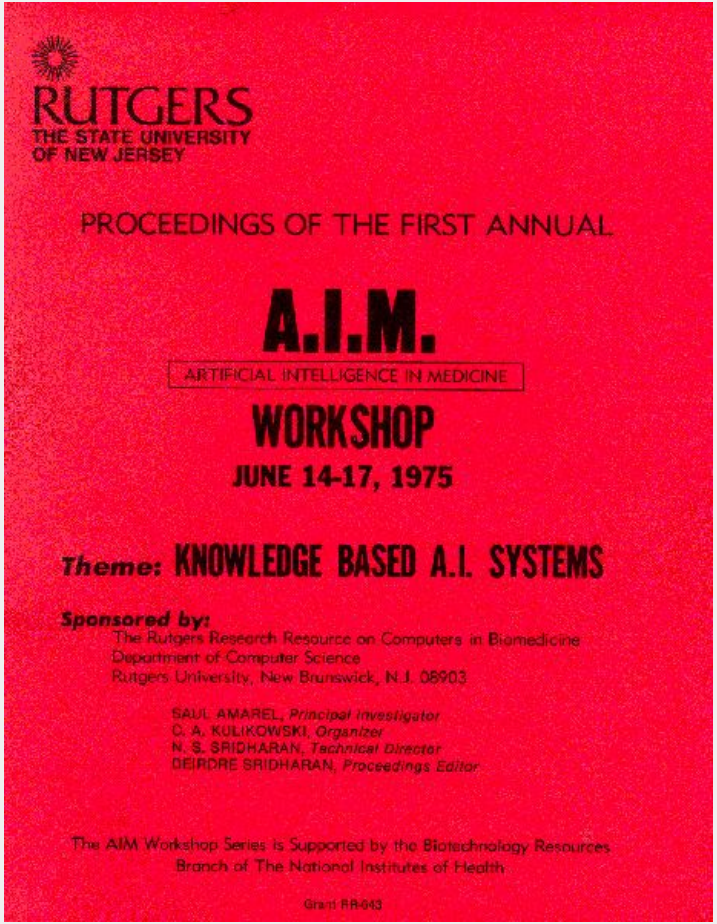
Digitizing Health Data 1950s

- While the AI field was moving along with experiments with robots (Shakey) and chatbot (ELIZA), medicine was slower to respond to automation
- The medical field began with digitization of health data
- Medical Literature and Analysis and Retrieval System MEDLARS; online counterpart: MEDLINE, 30 million references to journal articles in life sciences with a concentration on biomedicine, 1789 to present)
- and search engines of medical records (PubMed by National Library of Medicine)



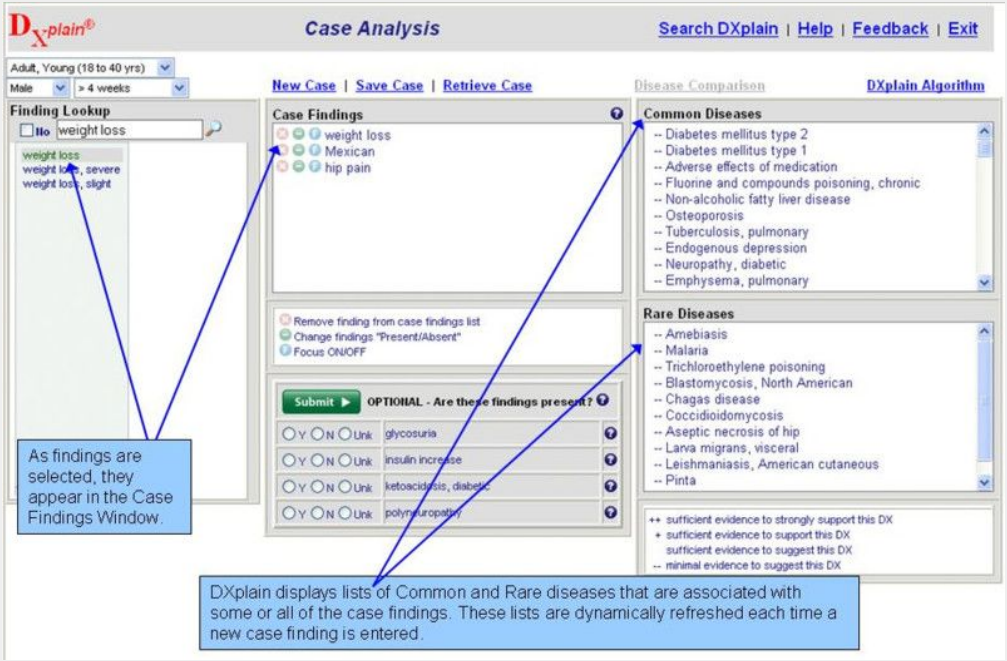
First AI Conferences 1970s

- Researchers in medicine worked on information sharing and networking systems for medicine.
- In 1971, The Research Resource on Computers in Biomedicine by Saul Amarel at Rutgers University. The Stanford University Medical Experimental – AI in Medicine created in 1973.
- The first AI in medicine workshop was held in 1975 at Rutgers University hotels by the National Institute of Health.



Medical Expert Systems 1970s

- In the 1970s these groups worked on expert system prototypes for applying AI to medical consulting.
- 1976 CASNET: model-building, consultation, database all brought together to apply information about diseases to individual patients and give physicians advice on patient management.
- 1970s MYCIN: backward chaining AI system with about 600 rules. Physicians could input patient information to get back list of potential bacterial pathogens and antibiotic recommendations.
- 1986 DXplain: input symptoms to get back different diagnoses. Served as an electronic medical textbook with diseases and references. At first release could handle 500 diseases.



Medical Chatbots 2010s

- Pharmabot developed in 2015 to assist in medication education for pediatric patients.
- Mandy was created in 2017 as an automated patient intake process for a primary care practice.

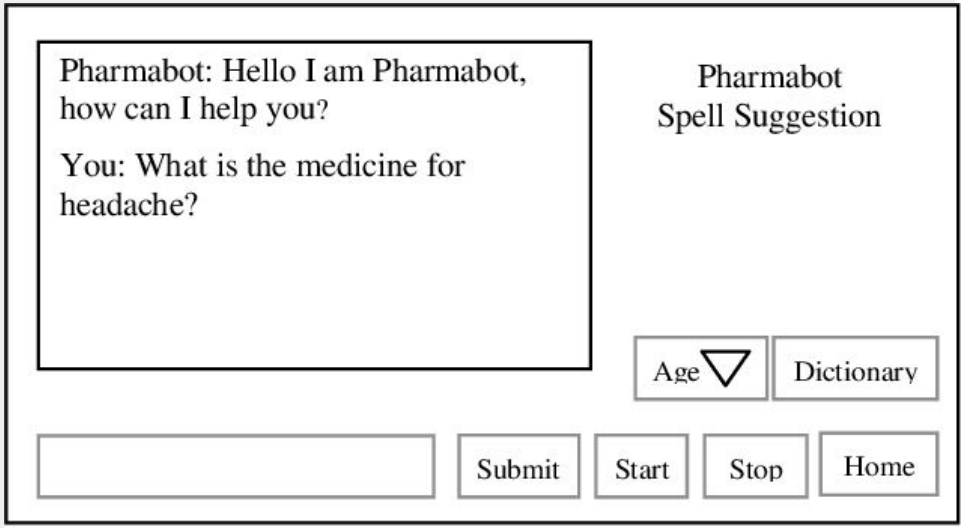


Figure 3. Conversation window

MANDY: Towards A Smart Primary Care Chatbot Application

Lin Ni[1], Chenhao Lu, Niu Liu, and Jiamou Liu[2]

Department of Computer Science, The University of Auckland, New Zealand
[1]lni600@aucklanduni.ac.nz, [2]jiamou.liu@auckland.ac.nz

Abstract. The paper reports on a proof-of-concept of Mandy, a primary care chatbot system created to assist healthcare staffs by automating the patient intake process. The chatbot interacts with a patient by carrying out an interview, understanding their chief complaints in natural language, and submitting reports to the doctors for further analysis. The system provides a mobile-app front end for the patients, a diagnostic unit, and a doctor's interface for accessing patient records. The diagnostic unit consists of three main modules: An analysis engine for understanding patients symptom descriptions, a symptom-to-cause mapper for reasoning about potential causes, and a question generator for deriving further interview questions. The system combines data-driven natural language processing capability with knowledge-driven diagnostic capability. We evaluate our proof-of-concept on benchmark case studies and compare the system with existing medical chatbots.

Keywords: Medicare chatbot; Patient interview; Natural language processing; AI and healthcare

Acknowledgement: The first author is partially funded by a scholarship offered by Precision Driven Health in New Zealand, a public-private research partnership aimed at improving health outcomes through data science. Initial progress of the research was reported in the PDH & Orion Health Blog https://orionhealth.com/global/knowledge-hub/blogs/meet-mandy-an-intelligent-and-interactive-medicare-system/

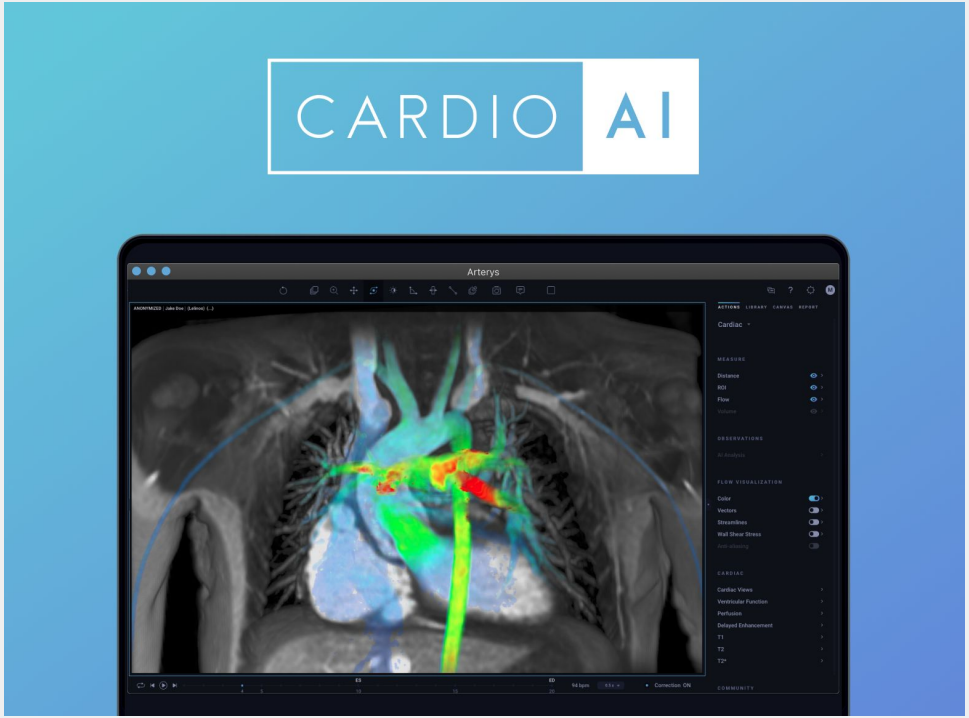
1 Introduction

Patients arriving at a primary care service sometimes need to wait for a long time before being advised by a doctor [26]. This is often due to high workload and limited resources at the primary care service [7]. To facilitate the process, nurses and other health care staffs usually take the role of patient intake. An incoming patient would be first greeted by a receptionist who carries out an intake inquiry. The receptionist would typically be someone who has a certain level of medical proficiency, and the inquiry involves collecting patient information and understanding the symptoms of the patient. A brief report is generated

Deep Learning in Medicine 2010s

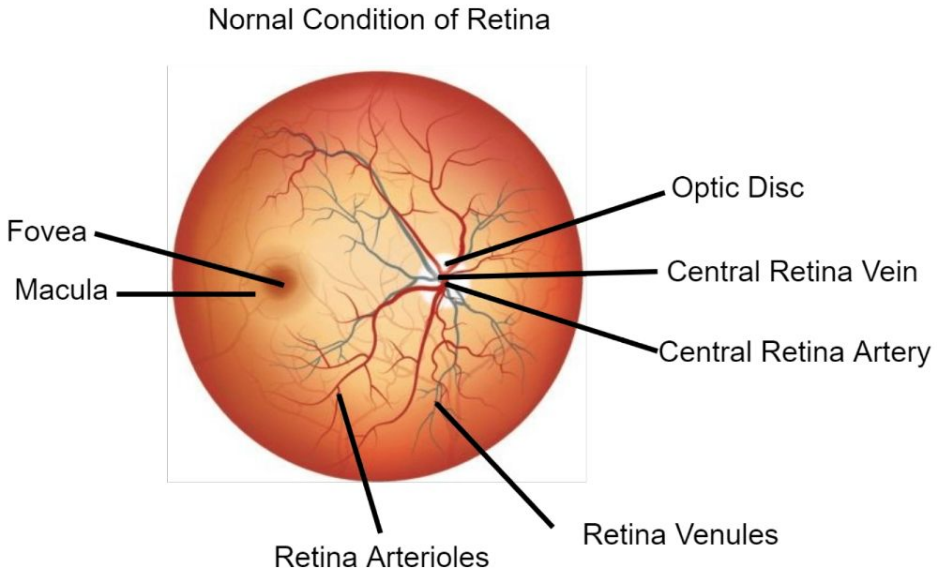
- In the 2010s researchers began to apply deep learning to medical imaging.
- Arterys was the first US FDA approved clinical DL health care product in 2017.
- CardioAI was able to analyze cardiac MRIs in seconds to give information about cardiac ejection fraction (now includes liver and lung imaging, chest and musculoskeletal xray images)

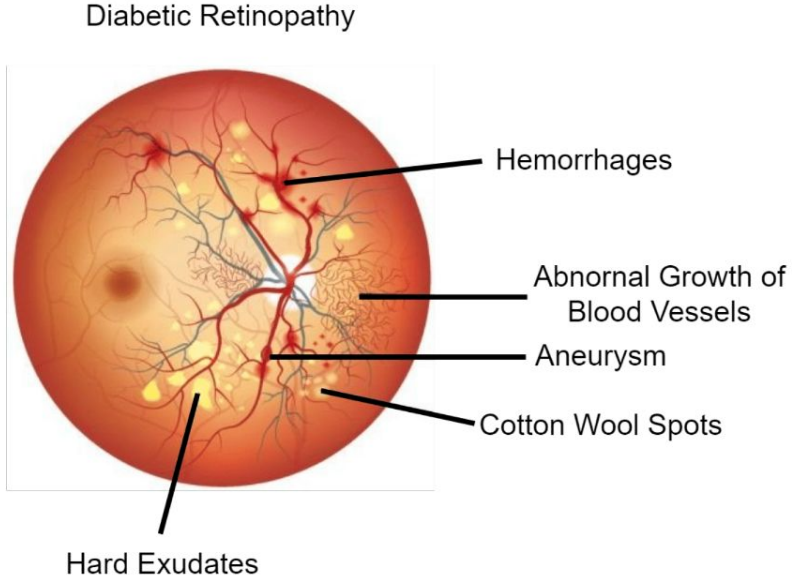
ARTERYS



Deep Learning in Medicine 2010s

- Vision for lesion detection, cardiovascular risk, progression of Alzheimers, and screen for diabetic retinopathy, etc.

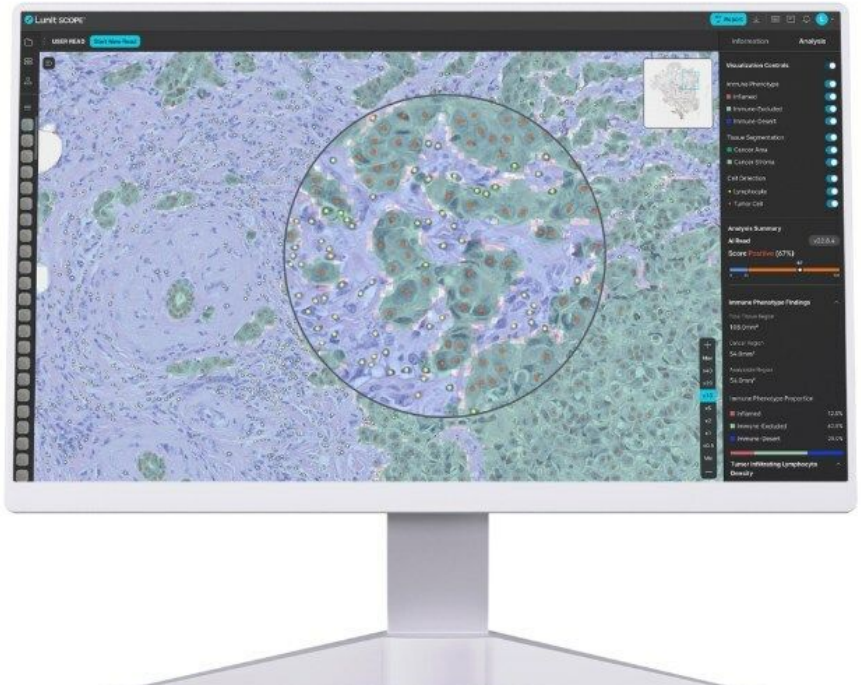


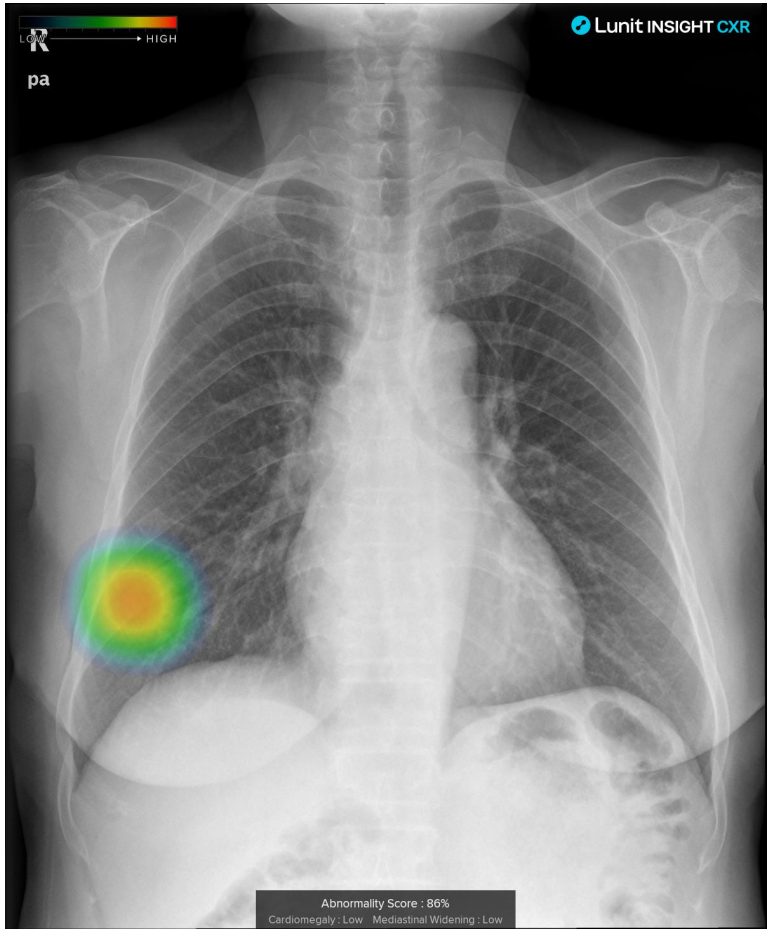


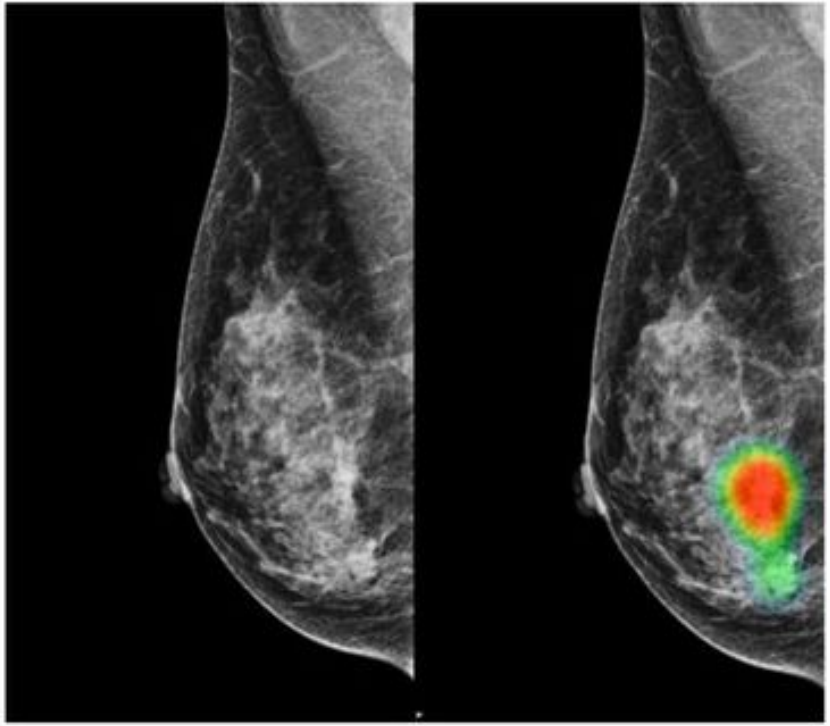
Computer Vision for Cancer Detection 2010s

- Computer vision has recently been used for cancer detection – Lunit is a South Korean company that has been successful with Lung and Breast cancer detection.

Lunit



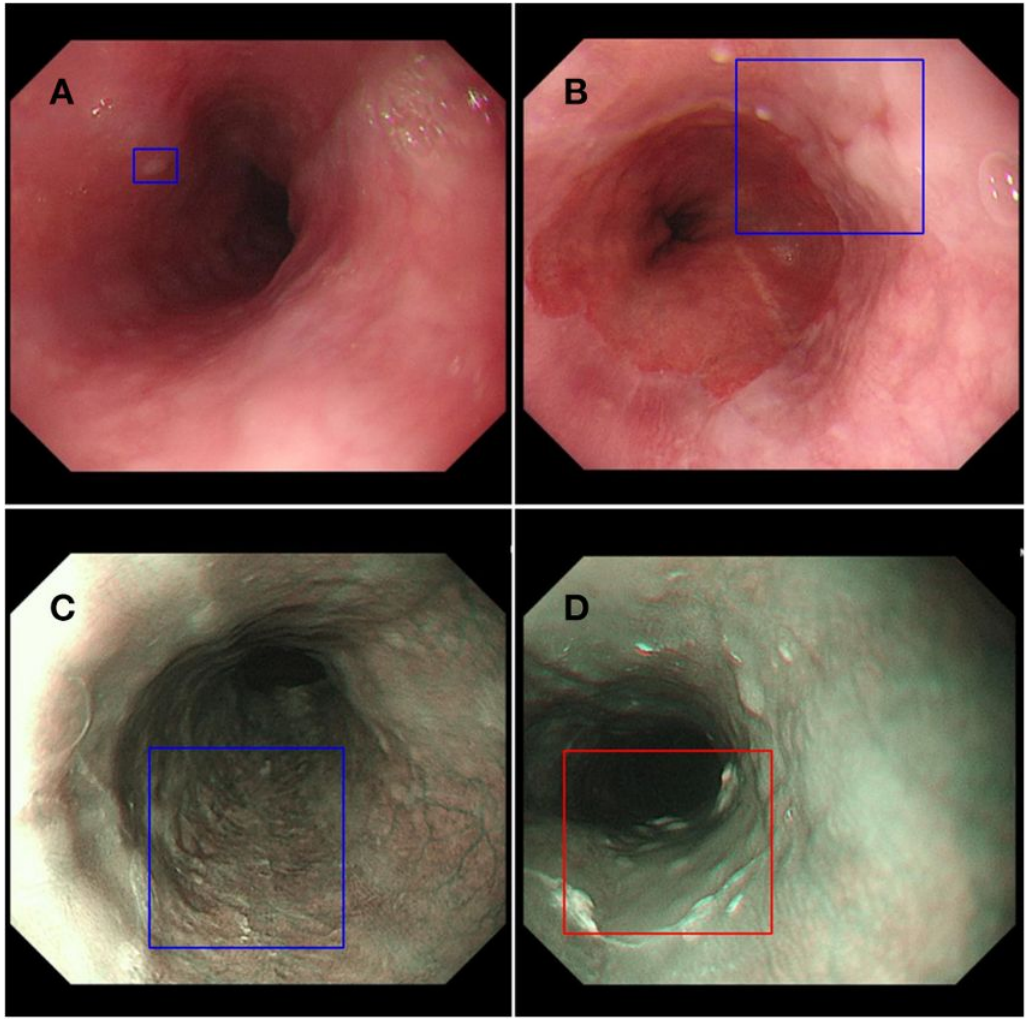

LOW
HIGH
R
pa
Lunit INSIGHT CXR
Abnormality Score : 86%
Cardiomegaly : Low
Mediastinal Widening : Low



Computer Vision for Cancer Detection 2010s

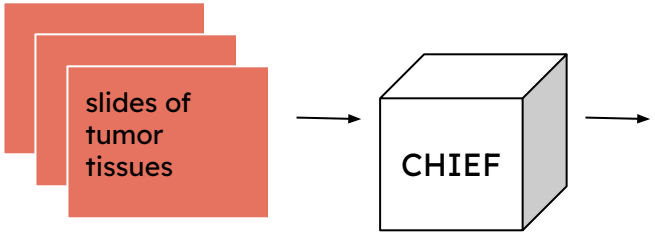
- Gastroenterology has been using AI for colonoscopies. It has been successful in identifying malignant/benign colon polyps and pancreatitis/pancreatic cancer.




A
B
C
D


ENDOANGEL
ENDOANGEL®
内镜精灵

More general: AI for Cancer Treatment



- 19 types of cancer cell detection
- tumor origin identification
- predicting patient outcomes
- identifying the presence of genes and DNA patterns related to treatment response in tumor micro-environment

nature

Explore content | About the journal | Publish with us | Subscribe

nature > articles > article

Article | Published: 04 September 2024

A pathology foundation model for cancer diagnosis and prognosis prediction

Xiyue Wang, Junhan Zhao, Eliana Marostica, Wei Yuan, Jietian Jin, Jiayu Zhang, Ruijiang Li, Hongping Tang, Kanran Wang, Yu Li, Fang Wang, Yulong Peng, Junyou Zhu, Jing Zhang, Christopher R. Jackson, Jun Zhang, Deborah Dillon, Nancy U. Lin, Lynette Sholl, Thomas Denize, David Meredith, Keith L. Ligon, Sabina Signoretti, Shuji Ogino, ... Kun-Hsing Yu + Show authors

Nature **634**, 970–978 (2024) | Cite this article

43k Accesses | **366** Altmetric | Metrics

Abstract

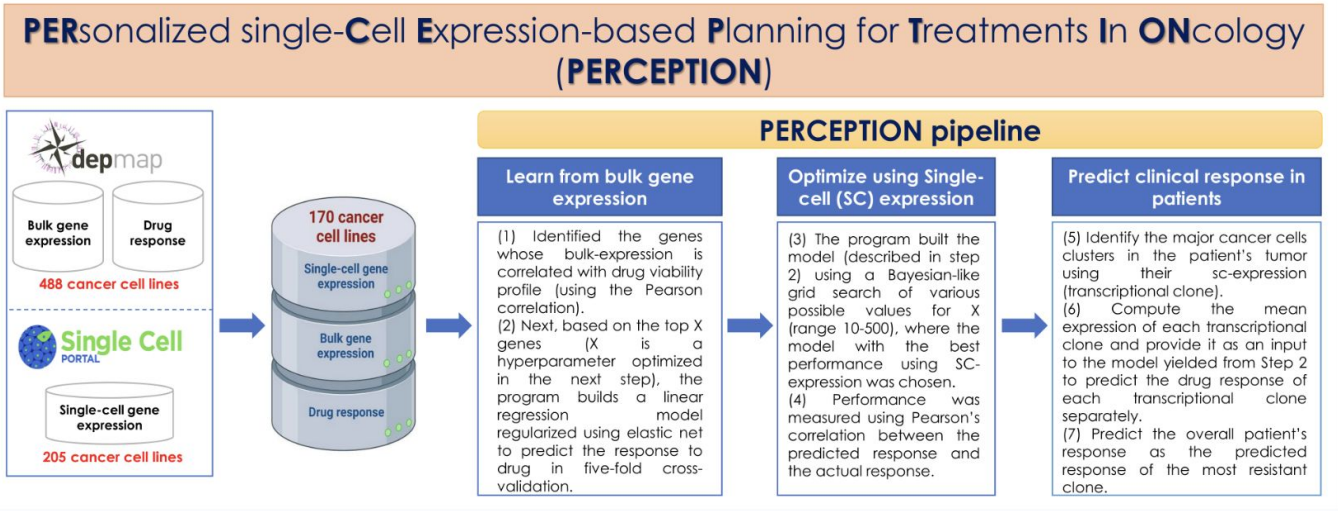
Histopathology image evaluation is indispensable for cancer diagnoses and subtype classification. Standard artificial intelligence methods for histopathology image analyses have focused on optimizing specialized models for each diagnostic task[1,2]. Although such methods have achieved some success, they often have limited generalizability to images generated by different digitization protocols or samples collected from different populations[3]. Here, to address this challenge, we devised the Clinical Histopathology Imaging Evaluation Foundation (CHIEF) model, a general-purpose weakly supervised machine

CHIEF Model 2024 for

More specific: Single-cell RNA Sequencing

Bulk sequencing cancer DNA & RNA
- average of all tumor cells
- difficult to target tumors

Single cell RNA sequencing
- higher resolution diagnosis
- individual cell level targetted
- more lasting drug responses

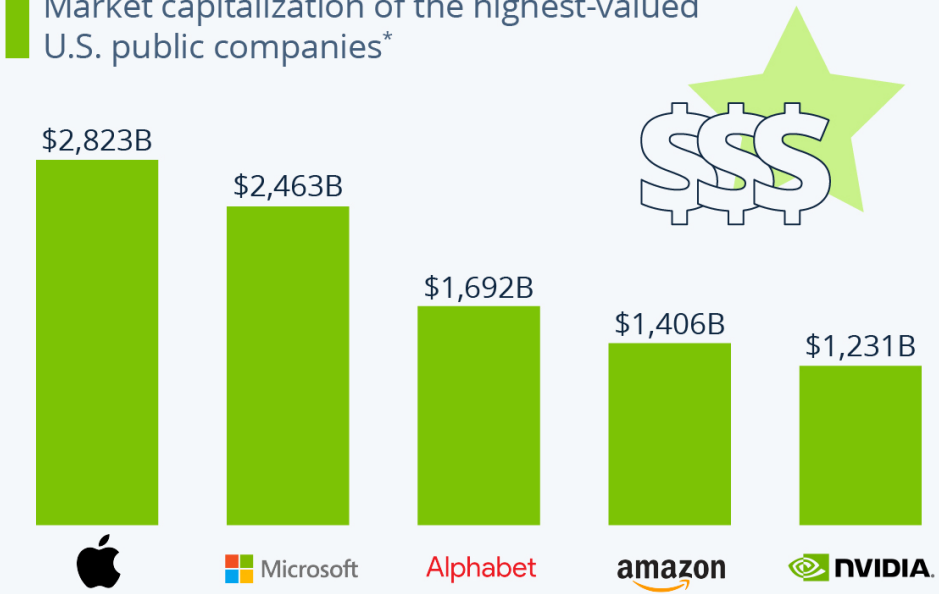


This week!

- Why is hardware so important for AI?
- How does labor and global economics play a role in modern AI growth?
- How do computing advances change AI accessibility and productizing?

Nvidia Cements Membership in Trillion Dollar Club

Market capitalization of the highest-valued U.S. public companies*



* As of market open on August 24, 2023

Source: Yahoo Finance

statista

SUPERCHARGED

A $40,000 Nvidia chip has become the world's most sought-after hardware

Companies and governments want to deploy generative AI—but first they need access to Nvidia's H100 chips

By **Michelle Cheng** Published August 18, 2023



Nvidia is the AI company of the moment. Photo: Ann Wang (Reuters)

It seems like everyone wants to get their hands on Nvidia's H100 chips these days.

Microsoft and Google, which are building out generative AI-driven search engines, are some of the biggest customers of Nvidia's H100 chips. Server manufacturers claim to have waited more than six months to receive their latest orders. Venture capital investors are buying up H100 chips for the startups they invest in.

RELATED CONTENT

GO MEDIA
AV CLUB DEADSPIN GIZMODO
the inventory JALOPNIK JEZEBEL
KOTAKU lifehacker the ONION
QUARTZ THE ROOT TAKEOUT

CHIPS

Nvidia Stock Rises. Why More Volatility Is Coming.

By George Glover Follow and Adam Clark Follow

Updated March 12, 2025, 5:09 pm EDT / Original March 12, 2025, 5:15 am EDT

Share Resize Reprints

In this article

↑ NVDA ↓ SPX ↓ 2330 ↓ AMD ↓ AVGO



Nvidia CEO Jensen Huang. (COURTESY NVIDIA)

Nvidia ↑ NVDA +0.64% stock rose Wednesday as it continued to pare back some of its recent losses—but more big swings could be ahead.

Shares in the chip maker rose 6.4% to $115.74, having closed the previous session up 1.7%. The benchmark S&P 500 ↓ SPX -1.22% index climbed 0.5%.

Nvidia is down 17% over the past month through Tuesday's close, tumbling as investors have fretted that U.S. President Donald Trump's tariff plans will drag down growth and trigger a flare-up of inflation, both of which could weigh on chip sales.

2020s: GPUs for AI

NVIDIA Blackwell GPUs Sold Out: Demand Surges, What's Next?

Published October 14, 2024 |  Written by **Esther Shein**

The supply of next-gen processors is sold out for the next 12 months as company demand for more sophisticated AI models and applications continues unabated.

Nvidia continues to ride the AI wave as the company sees unprecedented demand for its latest Blackwell GPU next-generation processors. The supply for the next 12 months is sold out, Nvidia CEO Jensen Huang told Morgan Stanley analysts during an investors' meeting.

A similar situation occurred with Hopper GPUs several quarters ago, Morgan Stanley Analyst Joe Moore pointed out.

Nvidia's traditional customers are driving the overwhelming demand for Blackwell GPUs, including major tech giants such as AWS, Google, Meta, Microsoft, Oracle, and CoreWeave. Every Blackwell GPU that Nvidia and its manufacturing partner TSMC can produce over the next four quarters has already been purchased by these companies.

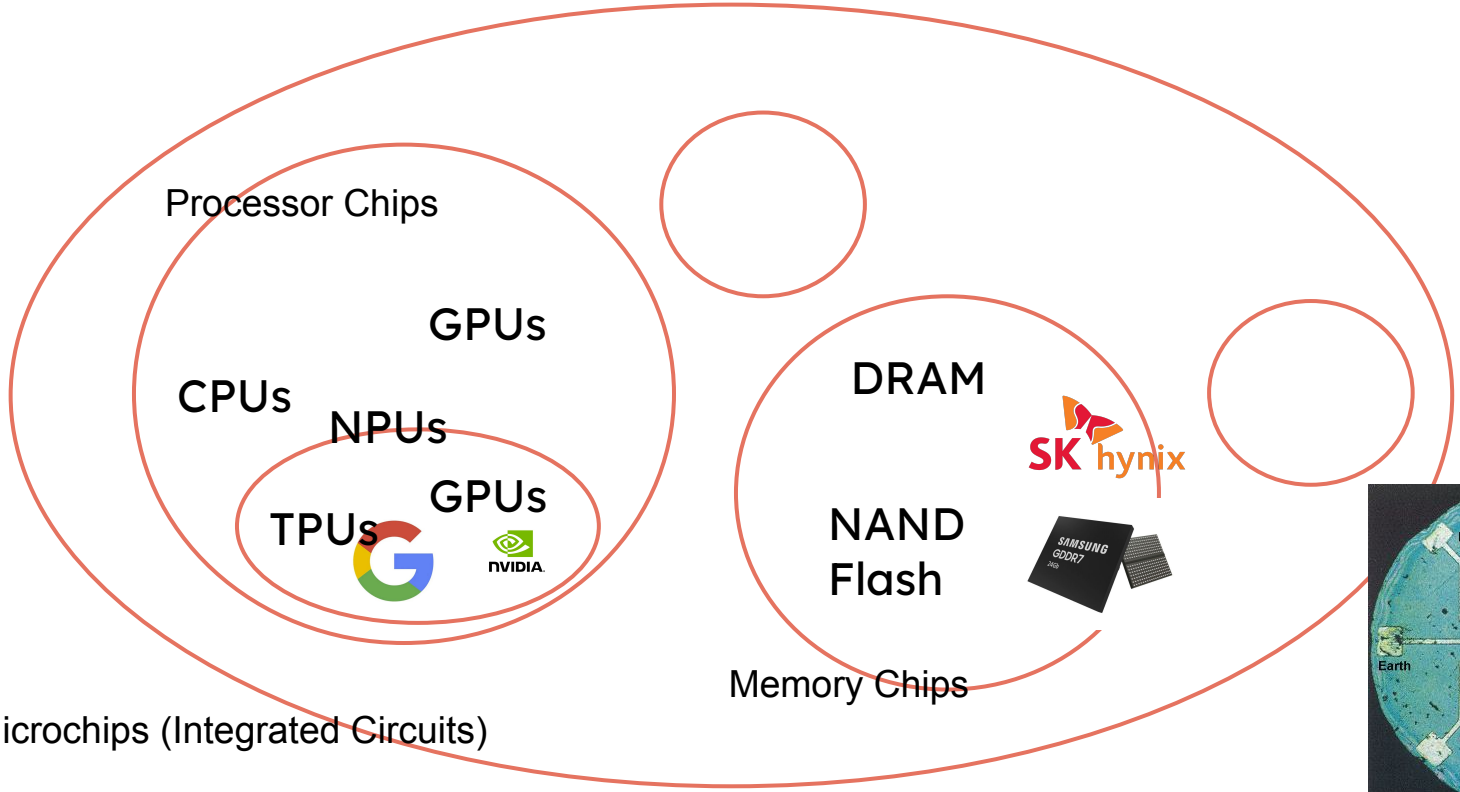
The AI Boom Is Giving Rise to "GPU-as-a-Service"

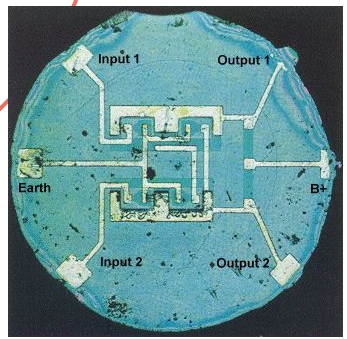
The industry harvests idle compute for AI startups that need it

BY JUAN PABLO PEREZ | 20 JAN 2025 | 3 MIN READ

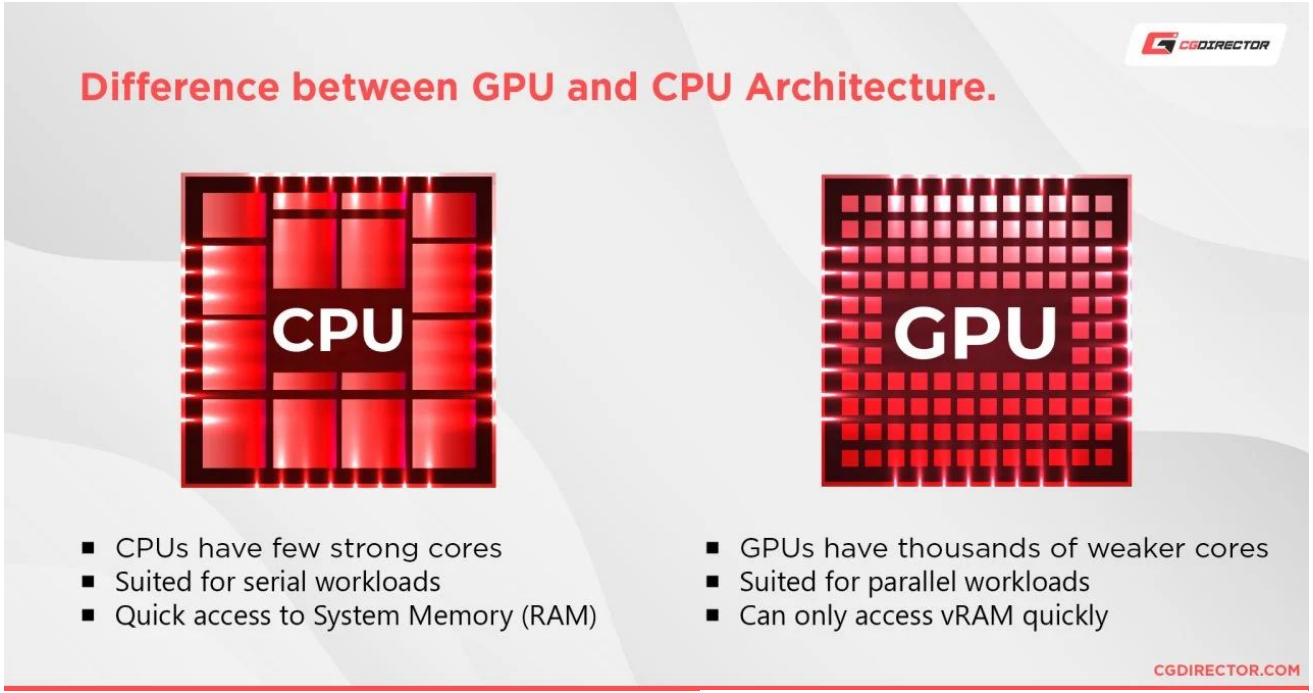


Microchips Family





Fairchild IC



GPUs

- better for batch processing
- graphics, deep learning (matrix multiplications)

CPUs

- typical PC microchips
- fast and good with general purpose computing

Chip Production

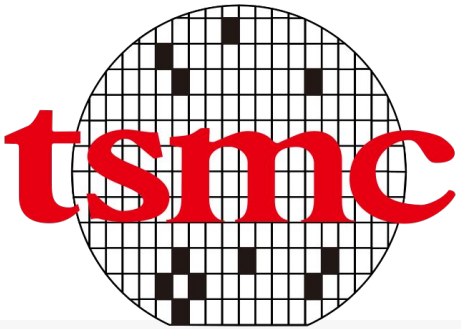
Lithography Machines

Canon

ASML

Fabrication Factories

intel



SAMSUNG

Chip Design + Software

NVIDIA

Qualcomm

Chip Production: Lithography

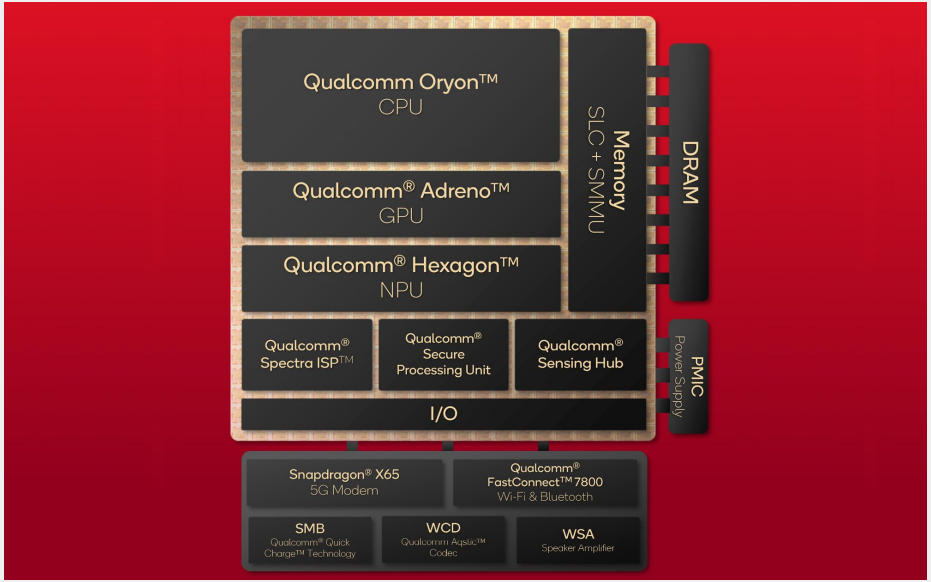


ASML EUV Lithography

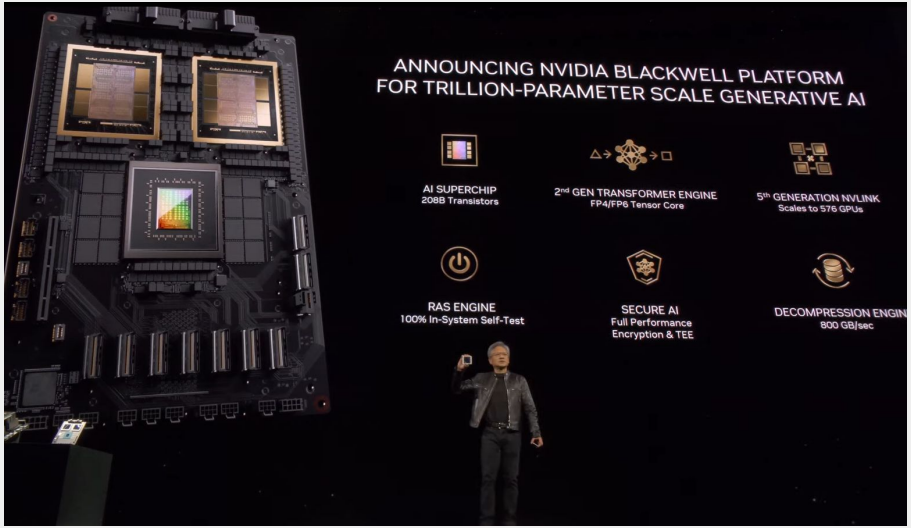


Canon DUV Lithography

Chip Production: Chip Design



Qualcomm Snapdragon SoC



Nvidia Blackwell Platform

Chip Production: Fabs & Foundries

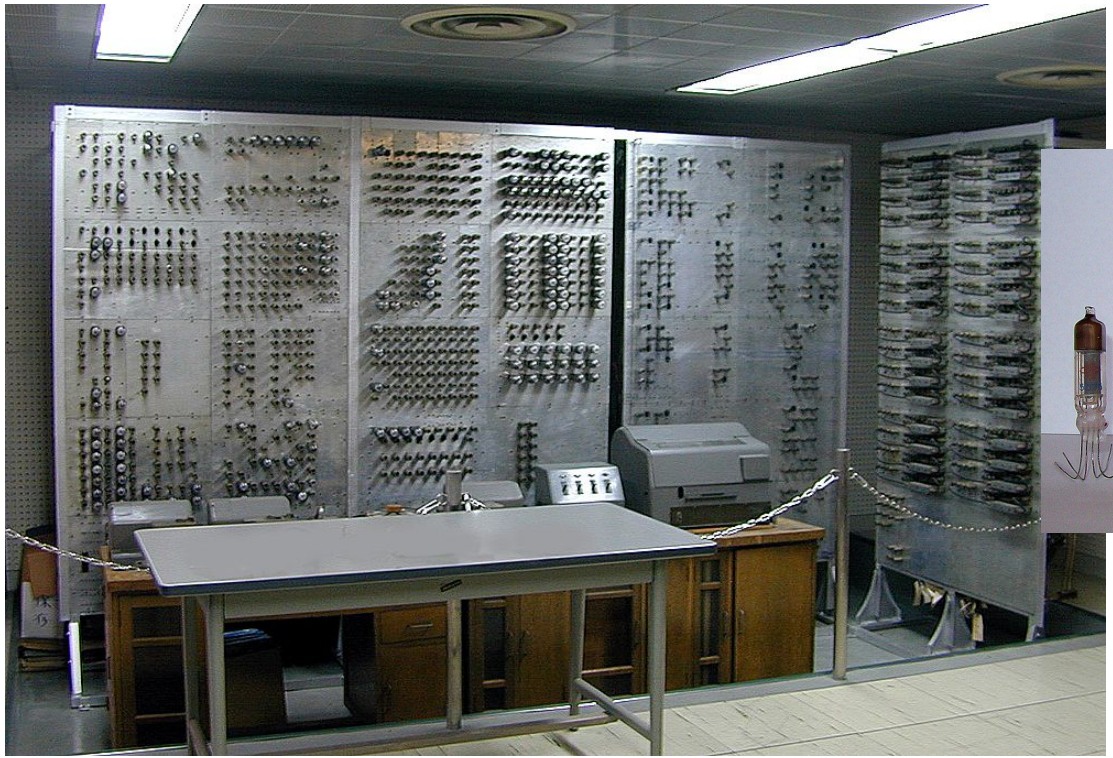


Samsung Pyeongtaek Fab



TSMC Foundry in Arizona

Beginning of semiconductor chips: Vacuum Tubes to Transistors (1940s)





ENIAC 1945

Beginning of semiconductor chips: Vacuum Tubes to Transistors (1940s)

Researchers at Bell Labs and AT&T start making semiconductors



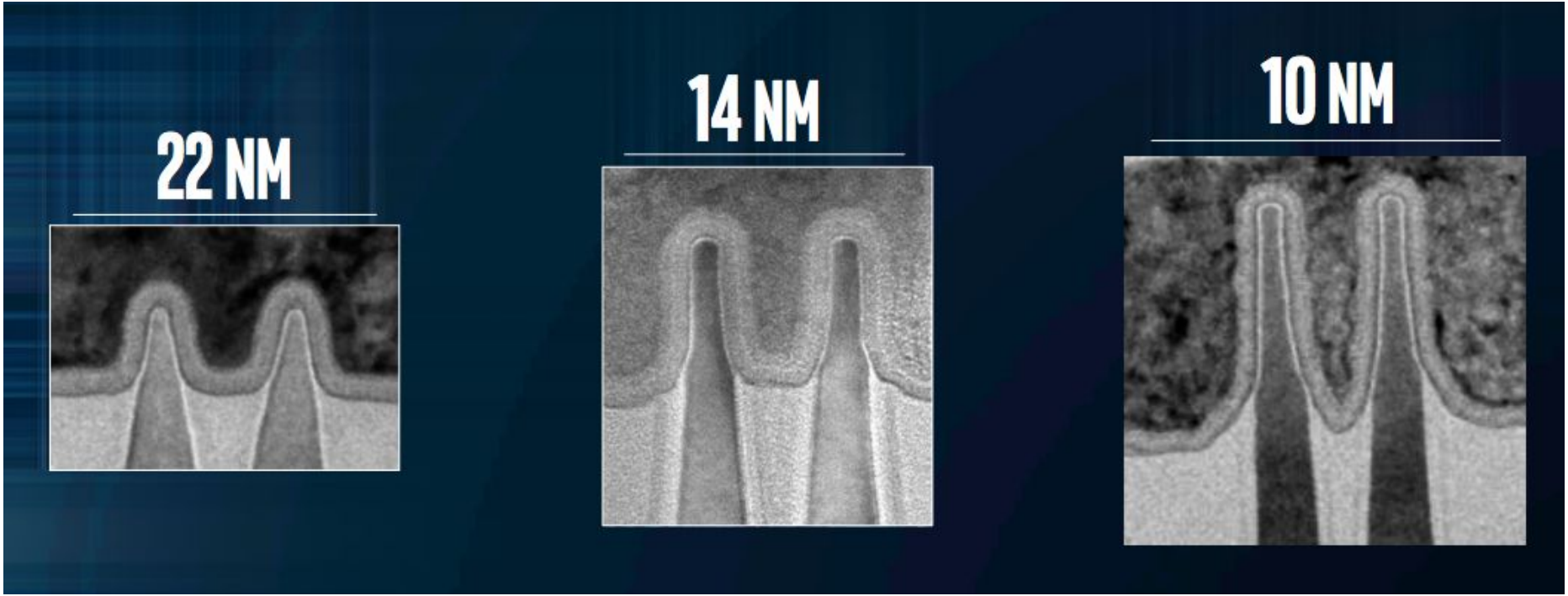
Silicon



Phosphorous

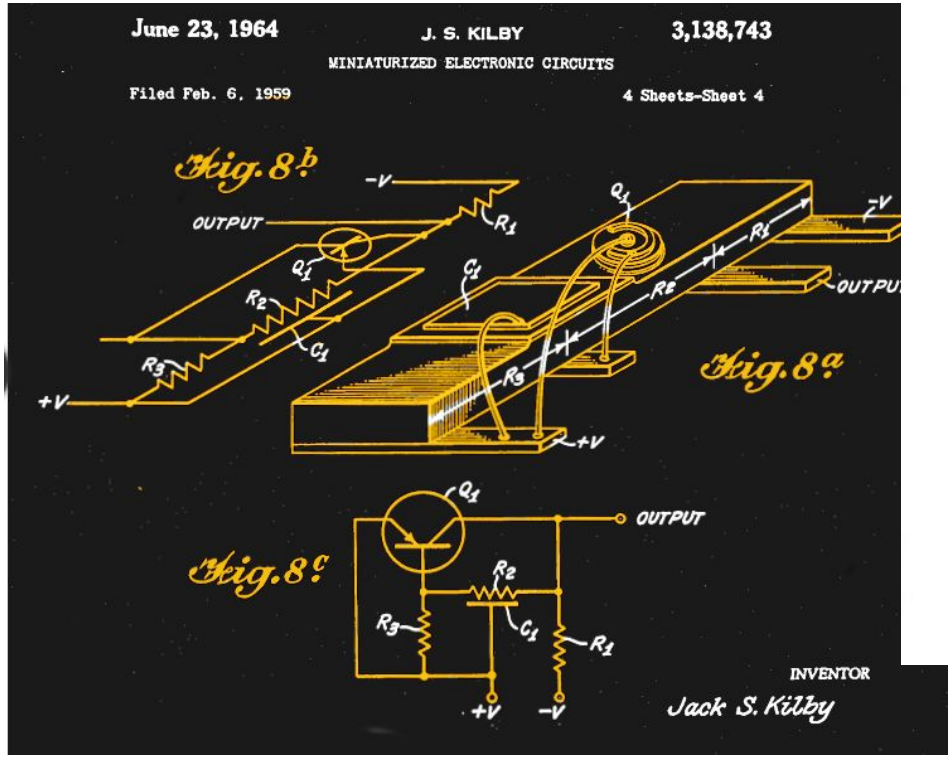
Beginning of semiconductor chips: Vacuum Tubes to Transistors (1940s)

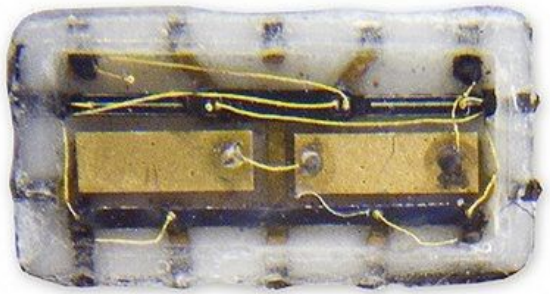
Each transistor is nanometers wide. Each chip carries billions of transistors



First Integrated Circuits 1958

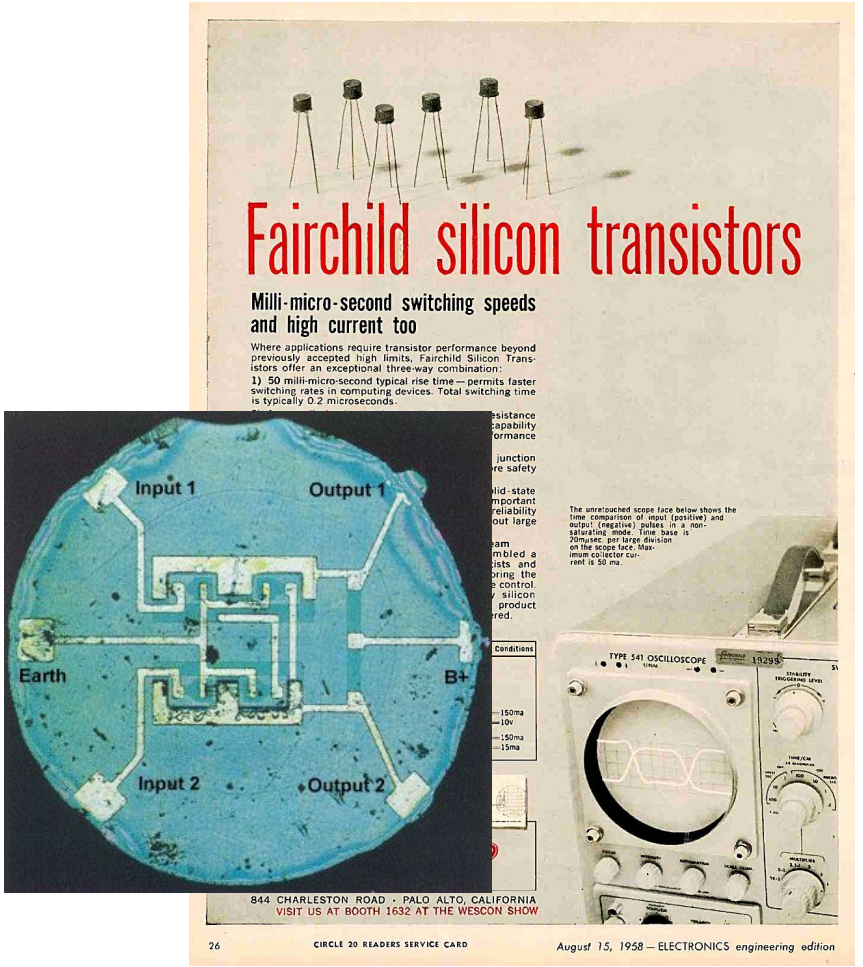
Kilby at Texas Instruments invents the Integrated Circuit





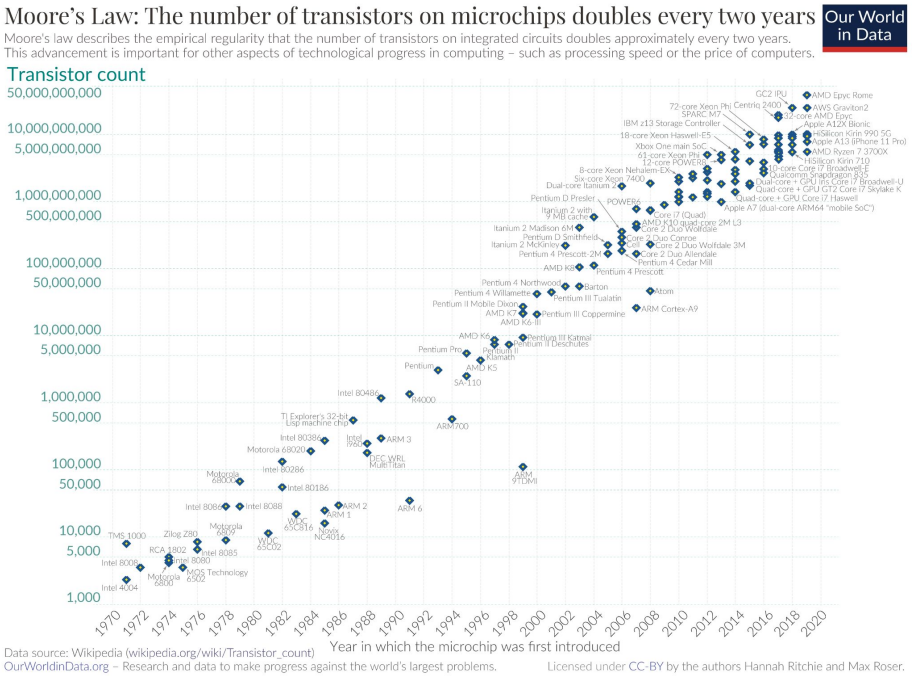
Fairchild Semiconductors

- Micrologic Chips 1958
- Apollo (NASA) program used Fairchild chips!
- Sales ballooned from $500,000 1958 to $21m two years later
- Founder Noyce started slashing prices to address wider consumer need ($120 → $15)
- 1965: military applications made up 95% of chips



Defense to Consumer Tech

- Cold War U.S. Defense using over 95% of chips up to the mid-1960s
- used in command, control technologies like satellites, sonar, torpedoes, telemetry systems
- …to the civilian sector (private market) in the late 1960s
- Moore's Law (1965) that every year for the next decade, the number of components possible to fit on silicon would double



Moore's Law (1965)

Chips come to Asia!

- U.S. starts investing in Japanese technology post WWII
- Morita founds Sony (Sonus + Sonny) 1946
- “licensing” strategy: Japanese electronics paid U.S. firms IP fees (4.5% to Fairchild, 3.5% to TI, 2% Western Electric)
- Japanese efficient manufacturing was an easy way to start embedding consumer products with chips (1958)



Sony Radios

Japan overtakes U.S.

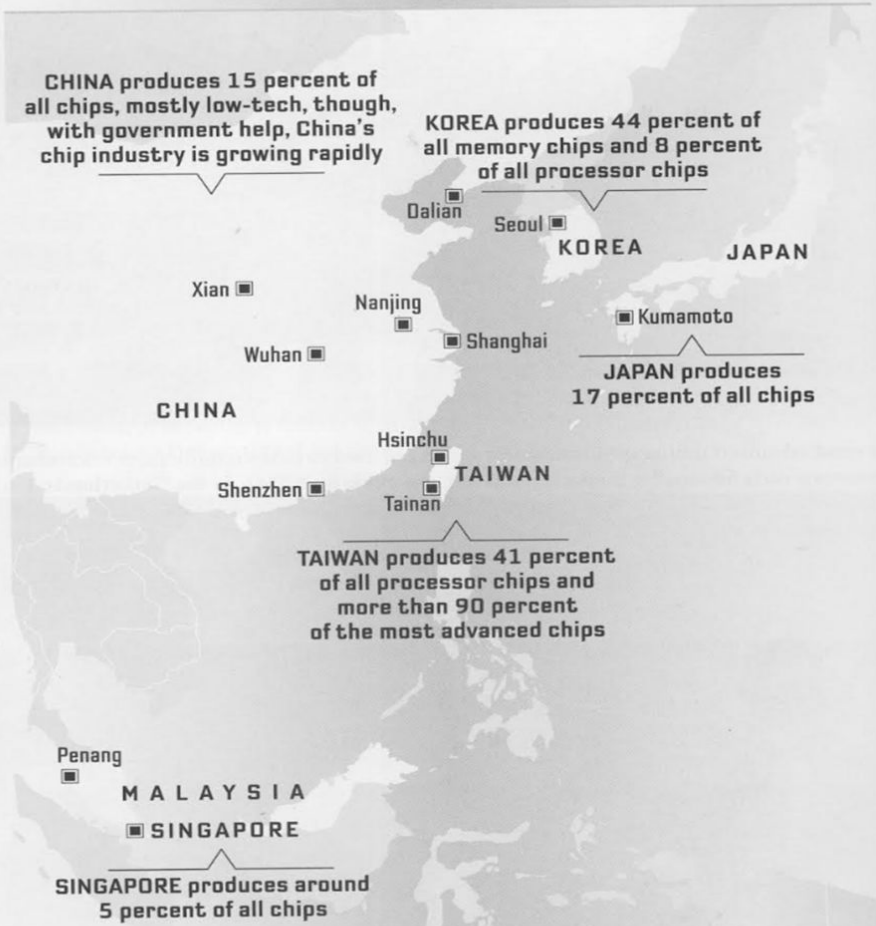
- By 1964 Japan overtakes U.S. in discrete transistor production
- U.S. was still producing the most advanced chips
- From 1965 to 1985 Japanese export of semiconductors and consumer products with them grew from $600 million to $60 billion



Sharp Calculator 1967

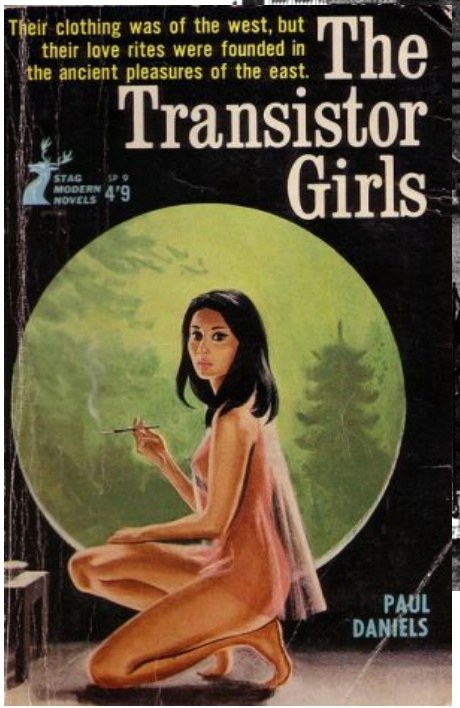
EAST ASIA PRODUCES:

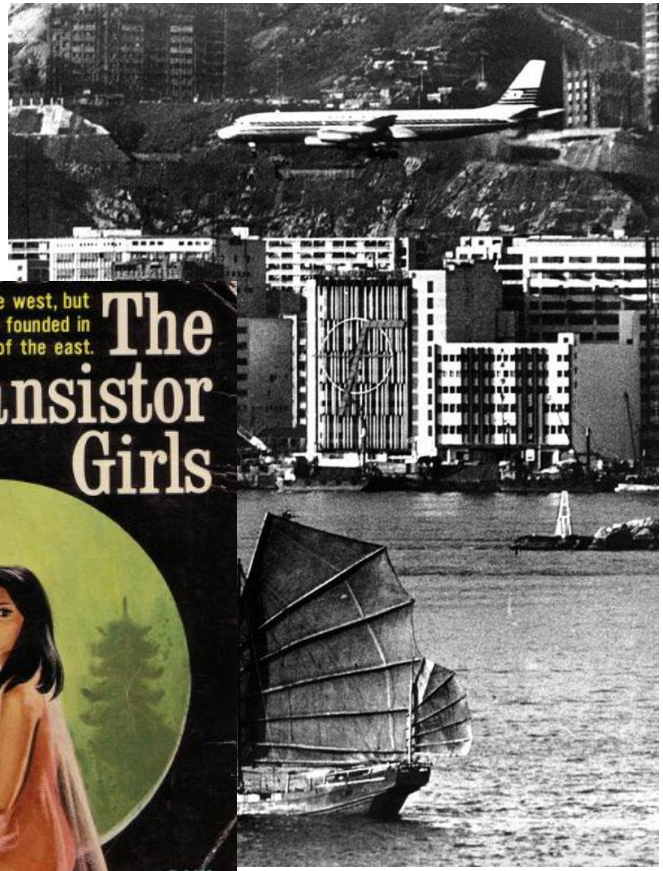
90 percent of all memory chips, 75 percent of all processor (logic) chips, and 80 percent of all silicon wafers



Cheaper labor in Asia 1960s

- Companies like Fairchild Semiconductor, Texas Instruments, Motorola start building factories in Hong Kong, Singapore, Malaysia
- Workers here were paid about 1/10 the wages of Americans
- HK 25c (/hour), Taiwan 19c, Malaysia 15c, Singaporean 11c, South Korean 10c
- Mostly women: “better on dexterity tests”





Fairchild on Kowloon

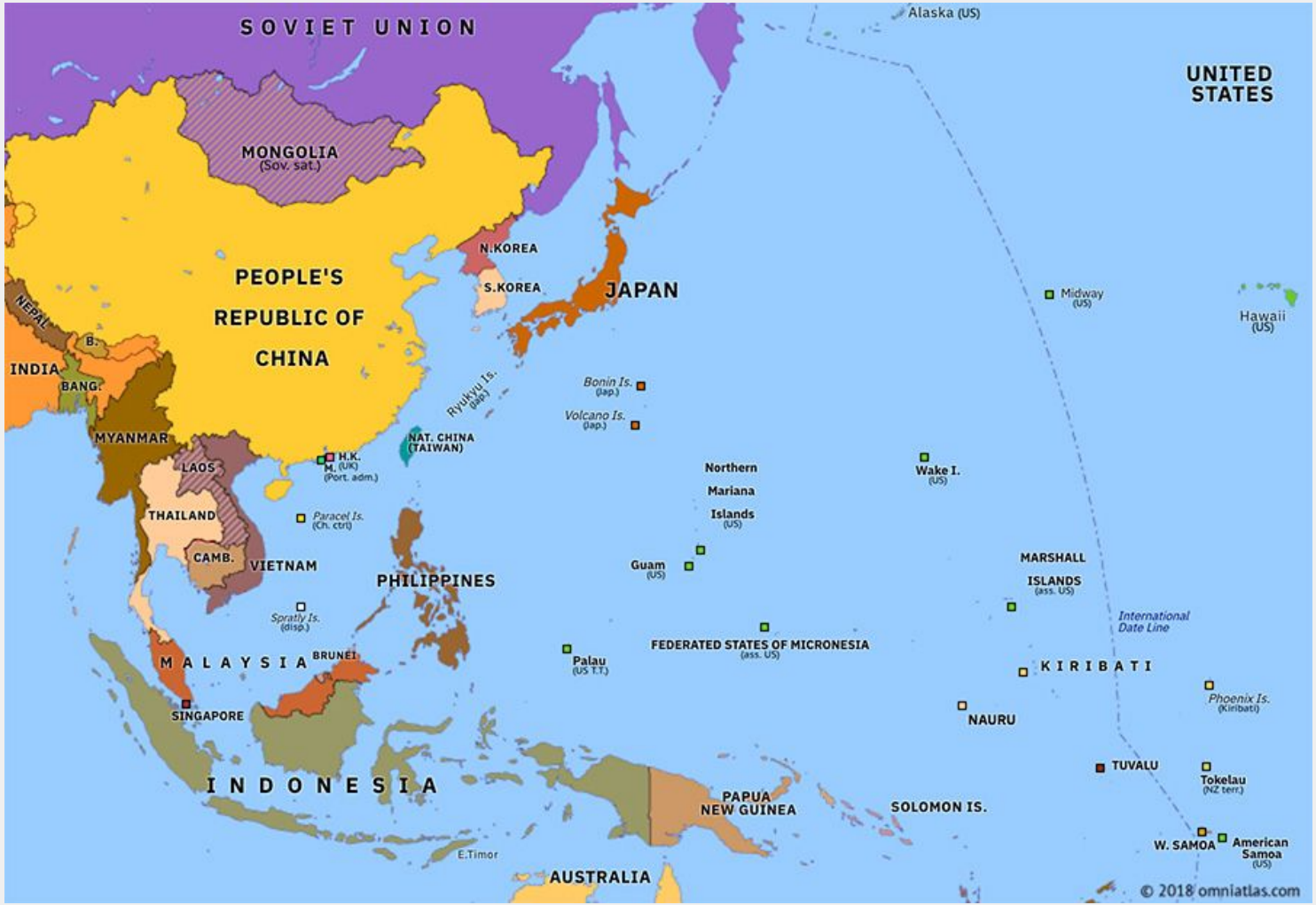
post-Vietnam War Political Landscape

- With the loss of the U.S. in the Vietnam War looming neighboring Asian countries look to U.S.-alliance
- China's first atomic weapons test 1964
- Taiwan in particular was concerned that the fall of South Vietnam would mean Communist strengthening in the region and a larger Communist Chinese threat.

- Taiwan turned to strategic alliance with the U.S. and partnered with Texas Instruments – first plant in 1969

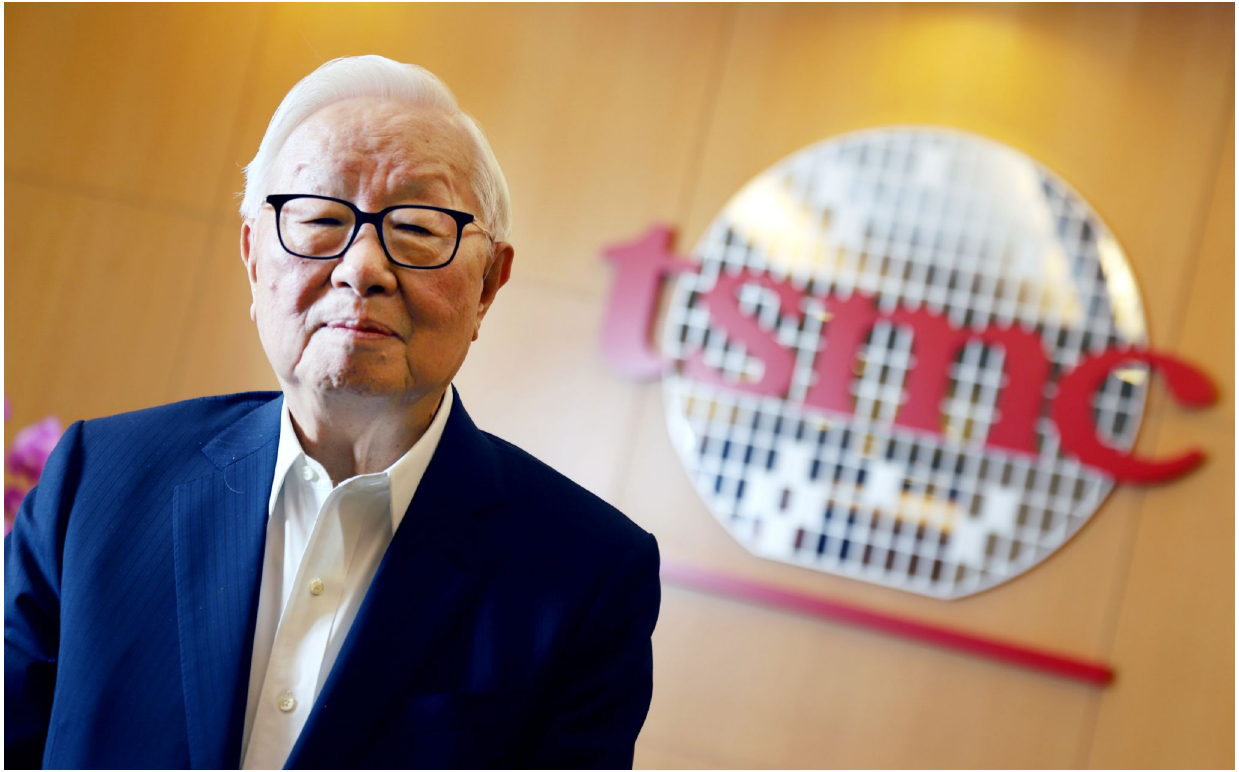


Minister K.T. Li


SOVIET UNION
Alaska (US)
UNITED STATES
MONGOLIA
(Sov. sat.)
PEOPLE'S REPUBLIC OF CHINA
N.KOREA
S.KOREA
JAPAN
Midway (US)
Hawaii (US)
NEPAL
B.
INDIA
BANG.
MYANMAR
Ryukyu Is. (jap.)
Bonin Is. (jap.)
Volcano Is. (jap.)
NAT. CHINA (TAIWAN)
H.K.
M. (UK)
(Port. adm.)
LAOS
THAILAND
Paracel Is. (Ch. ctrl)
CAMB.
VIETNAM
PHILIPPINES
Northern Mariana Islands (US)
Guam (US)
Wake I. (US)
MARSHALL ISLANDS (ass. US)
Spratly Is. (disp.)
FEDERATED STATES OF MICRONESIA (ass. US)
International Date Line
Palau (US T.T.)
MALAYSIA
BRUNEI
KIRIBATI
Phoenix Is. (Kiribati)
SINGAPORE
NAURU
TUVALU
Tokelau (NZ terr.)
INDONESIA
PAPUA NEW GUINEA
SOLOMON IS.
W. SAMOA
American Samoa (US)
E.Timor
AUSTRALIA
© 2018 omniatlas.com

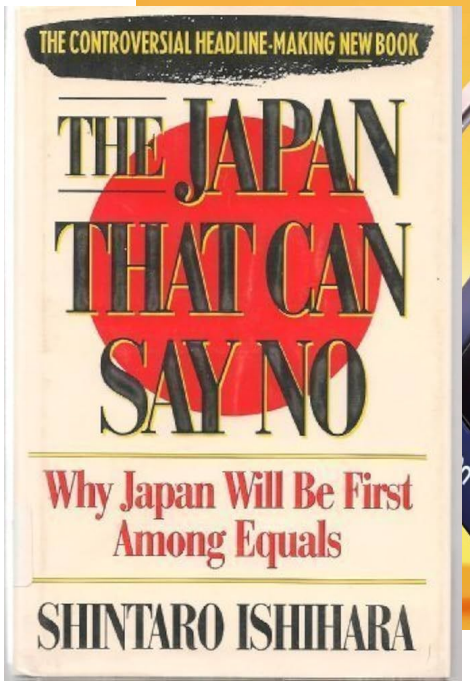
Moriss Chang to lead TSMC

- Minister Li hires Morris Chang to lead TSMC in 1985
- Blank cheque from Taiwanese government (48% share)
- TSMC becomes first “foundry” – “Gutenberg for semiconductors”
- Philips gave $58m, production technology for 27.5%



Japan's productivity threatens the U.S. 1980s

- Japan's high productivity starts to threaten U.S. dominance into the 1980s
- DRAM competition with Japan weakens U.S. memory chip production
- Morita: "Militarily we could never defeat the U.S. but economically we can overcome the U.S. and become number one in the world"
- The CIA coined this the "PAX JAPONICA"



coauthored with Morita 1989



Sony Walkman 1979

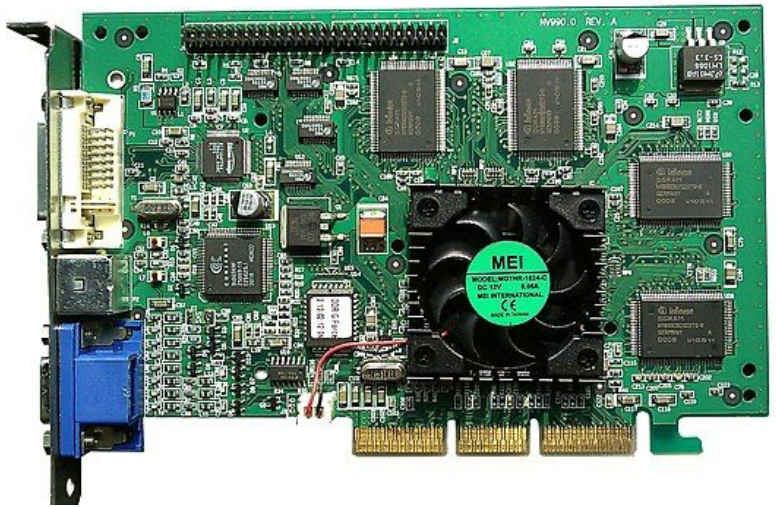
South Korean Hedge

- Silicon Valley looked to Korean producers to challenge Japanese threat
- Korean production even cheaper than Japan
- Several U.S. firms (e.g. Intel) began to sign joint ventures with Samsung in the 1980s
- Silicon Valley AMD CEO Jerry Sanders “enemy of my enemy is my friend”
- U.S. provided market and tech for Samsung chip production growth



Startups on Graphics Cards

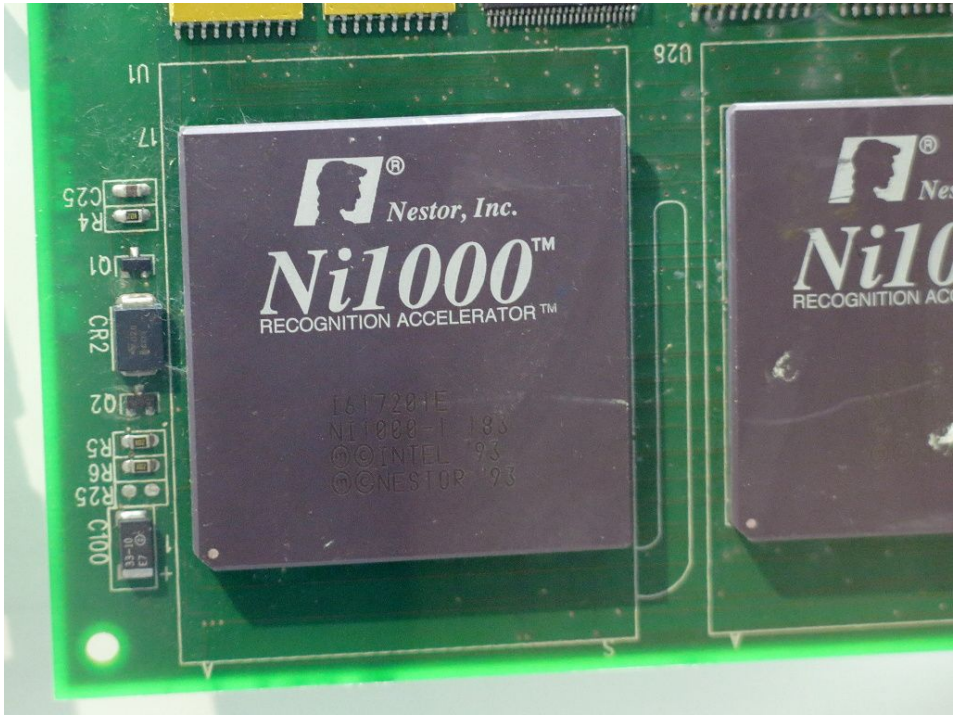
- In the 1980s Intel had a de facto monopoly on CPU microprocessors but graphics chips were a hot field for startups
- NVIDIA was founded in 1993 in Silicon Valley to design GPU chips which were mostly used for 3d graphics (for games and video)



1999 GeForce

Application Specific Integrated Circuits (ASICs)

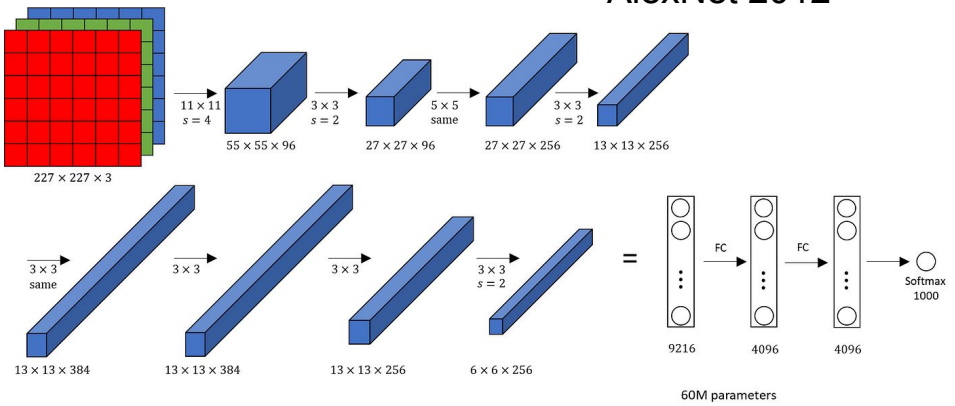
- 1980s Companies started developing microchips specifically made for neural networks
- 1980s Intel: ETANN 80170NX
- 1990s Nester/Intel: Nestor/Intel Ni1000 (DARPA) - could recognize handwritten characters
- Then 1990s, field-programmable gate arrays (FPGAs) for AI compute (away from ASICs)



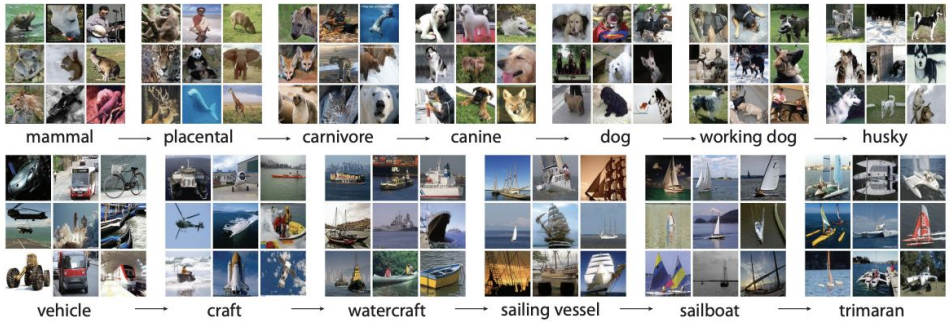
Nvidia's breakthrough? 2012 AlexNet

- AlexNet, a deep learning model proposed by researchers at U of Toronto, topped a ImageNet competition 2012 with 15% top-5 error (more than 10 percentage points lower than runner-up)
- Trained on 2 Nvidia GPUs!

AlexNet 2012



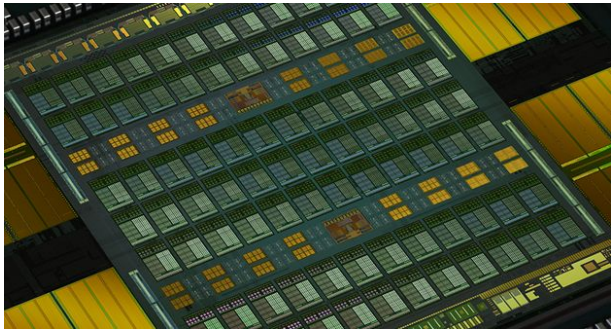
ImageNet 2009



Nvidia bets on AI future 2010 ~

- NVIDIA writes CUDA software that lets GPUS be programmed with standard programming languages (not needing graphics expertise). CUDA is only available on NVIDIA chips.
- Early 2010s Nvidia bets its future on AI hearing researchers are using GPUs not just for graphics
- NVIDIA has almost completed outsourced their manufacturing to TSMC and started focusing on software
- 2017 Tensor Cores specifically dedicated for AI/ML applications

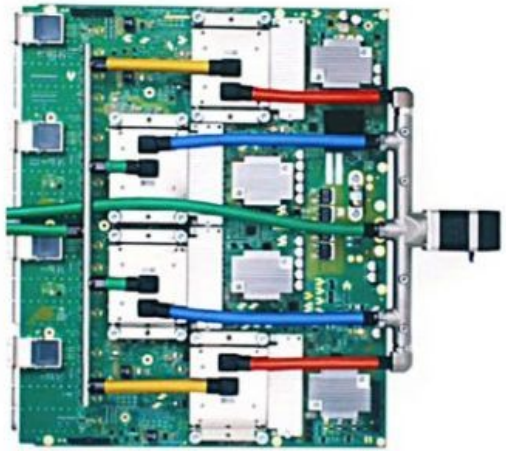




Tensor Cores in 2017 (Volta Architecture, V100 GPUs)

Big tech works on NPUs

- Other companies like Google also combined their software + hardware efforts for AI
- Google engineered the TPU (2016) to work well with Google's software Tensorflow (2015)
- Google TPU was used for AlphaGO 2016 showing magnitude higher efficiency over GPUs
- TPUs used first for Google search and Google StreetView

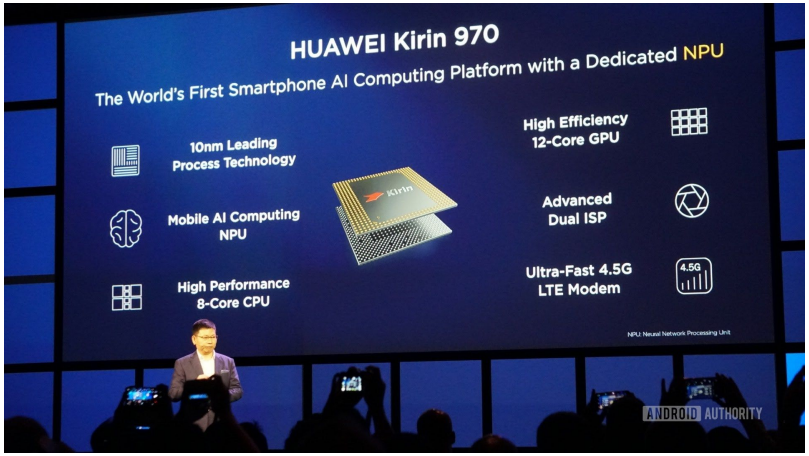


TPU V4



NPUs for Mobile

- 2015: Qualcomm's Snapdragon 820 is first NPU for mobile for on-device AI inference
- 2017: Huawei's Kirin 970, a NPU Neural Processing Unit, the first of its kind in a phone
- 2017: Apple's A11 Bionic chip introduced the *Apple Neural Engine*, a dual-core NPU capable of 600 billion operations per second, enabling real-time Face ID facial recognition and Animoji on the iPhone X



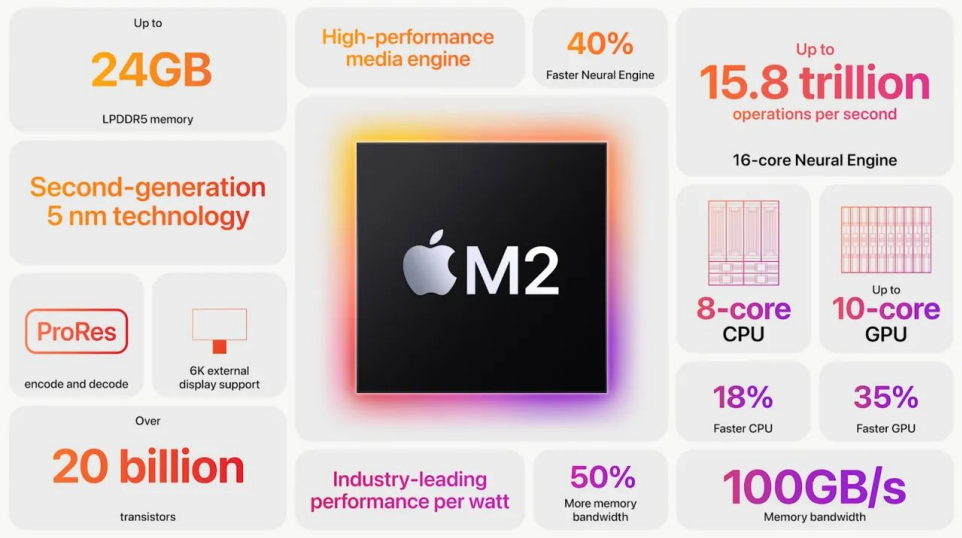
Huawei Kirin 970



Apple's Animoji

NPUs on the Cloud & Beyond

- 2019: Amazon introduced Inferentia chips for AI inference in its AWS cloud and Trainium for training
- Microsoft developing its own AI for Azure
- Google TPU tensorboards on GCP
- Apple's M1, M2 chips bring the Apple Neural Engine to the Macbook



Chip Acts around the world

- August 2022: U.S. CHIPS and Science Act: $52.7 billion in subsidies and tax credits for domestic semiconductor chip production
- February 2022: European Chips Act: seeks to double the EU's share of global semiconductor production to 20% by 2030 through a €43 billion investment
- 2014, China's "Big Fund": achieve self-sufficiency in semiconductor production, third phase launched in 2024, targeting a fundraising goal of approximately $47.5 billion



